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Review Article

Real-time semantic segmentation for autonomous driving: A review of CNNs, Transformers, and Beyond

Mohammed A.M. Elhassan^{a,b}, Changjun Zhou^{a,*}, Ali Khan^c, Amina Benabid^{a,b,d},
Abuzar B.M. Adam^e, Atif Mehmood^a, Naftaly Wambugu^f

^a School of Computer Science and Technology, Zhejiang Normal University, 688 Yingbin Road, Jinhua, 321004, Zhejiang, P.R. China

^b Zhejiang Institute of Photoelectronics & Zhejiang Institute for Advanced Light Source, Zhejiang Normal University, 688 Yingbin, P.R. China

^c Nanjing University of Aeronautics and Astronautics, Nanjing 211106, P.R. China

^d College of Mathematical Medicine, Zhejiang Normal University, 688 Yingbin Road, Jinhua, 321004, Zhejiang, P.R. China

^e Reliability and Trust (SnT), University of Luxembourg, 1855 Luxembourg City, Luxembourg

^f School of Architecture and Urban Planning, Shenzhen University, Shenzhen, P.R. China

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ABSTRACT

Real-time semantic segmentation is a crucial component of autonomous driving systems, where accurate and efficient scene interpretation is essential to ensure both safety and operational reliability. This review provides an in-depth analysis of state-of-the-art approaches in real-time semantic segmentation, with a particular focus on Convolutional Neural Networks (CNNs), Transformers, and hybrid models. We systematically evaluate these methods and benchmark their performance in terms of frames per second (FPS), memory consumption, and CPU runtime. Our analysis encompasses a wide range of architectures, highlighting their novel features and the inherent trade-offs between accuracy and computational efficiency. Additionally, we identify emerging trends, and propose future directions to advance the field. This work aims to serve as a valuable resource for both researchers and practitioners in autonomous driving, providing a clear roadmap for future developments in real-time semantic segmentation. More resources and updates can be found at our GitHub repository: <https://github.com/mohamedac29/Real-time-Semantic-Segmentation-Survey>

Contents

1. Introduction	2
1.1. Problem definition.....	4
1.2. Our contribution	4
1.3. Scope of the study.....	4
1.4. Paper structure.....	4
2. Methods: A survey.....	4
2.1. Model compression techniques	5
2.1.1. Network pruning.....	5
2.1.2. Neural architecture search (NAS).....	5
2.1.3. Knowledge distillation	5
2.1.4. Quantization	5

* Corresponding author at: School of Computer Science and Technology, Zhejiang Normal University, 688 Yingbin Road, Jinhua, 321004, Zhejiang, P.R. China.

E-mail addresses: mohammedac29@zjnu.edu.cn (M.A.M. Elhassan), zhouchangjun@zjnu.edu.cn (C. Zhou), alikh@nuaa.edu.cn (A. Khan), amina.benabid@zjnu.edu.cn (A. Benabid), abuzar.babikir@uni.lu (A.B.M. Adam), atifmeh@zjnu.edu.cn (A. Mehmood), wambugunaph@gmail.com (N. Wambugu).

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2.2.	Efficient CNN-based networks.....	5
2.3.	Efficient transformer-based networks.....	6
3.	Methods categorization	6
3.1.	Semantic segmentation based CNN networks	6
3.1.1.	Encoder-decoder based networks	6
3.1.2.	Two-branch networks	6
3.1.3.	Multi-branch networks.....	7
3.2.	Semantic segmentation based transformer networks	7
4.	Datasets and metrics for semantic segmentation	10
4.1.	Datasets.....	10
4.1.1.	ADE20K dataset	10
4.1.2.	Cityscapes dataset	10
4.1.3.	Camvid dataset	10
4.1.4.	PASCAL VOC 2012 dataset.....	10
4.1.5.	PASCAL context dataset.....	11
4.1.6.	COCOStuff-10K dataset.....	11
4.2.	Evaluation metrics.....	11
4.2.1.	Metrics for accuracy	11
4.2.2.	Computational efficiency metrics	11
5.	Benchmark analysis.....	12
5.1.	Benchmark results on cityscapes dataset.....	12
5.2.	Benchmark results on Camvid dataset	14
5.3.	Benchmark results on ADE20K dataset	14
5.4.	Benchmark results on COCOStuff dataset.....	14
6.	Challenges and research directions	16
6.1.	Key challenges in semantic segmentation	16
6.1.1.	The trade-off between accuracy and efficiency.....	16
6.1.2.	The challenge of high-quality training data	16
6.1.3.	Addressing domain gaps across datasets.....	16
6.2.	Future directions for semantic segmentation.....	16
6.2.1.	Harnessing transformer for real-time segmentation.....	16
6.2.2.	Expanding beyond traditional datasets	16
6.2.3.	Optimizing real-time segmentation for edge and mobile devices.....	16
6.2.4.	Knowledge distillation from large models to lightweight networks	16
6.2.5.	Scalability of real-time segmentation models	17
6.2.6.	Robustness to adverse conditions	17
6.2.7.	Energy-efficient inference.....	17
6.2.8.	Multi-modal segmentation.....	17
6.2.9.	Ethical and bias considerations.....	17
7.	Conclusion	17
	CRedit authorship contribution statement	18
	Declaration of competing interest.....	18
	Acknowledgment	18
	References.....	18

1. Introduction

In recent years, the rapid development and integration of big data, cloud computing, artificial intelligence, and related fields have brought concepts like intelligent transportation and smart cities into the spotlight. As people increasingly desire a smarter, more convenient, and better-quality life, the academic and commercial interest in these areas has surged. Recognizing the significant academic value and vast business potential, numerous academic institutions and industries have invested in scientific research aimed at enhancing human life through technology. Machine learning (ML), a crucial subset of artificial intelligence, is the driving force behind the new industrial revolution. It focuses on developing algorithms that can automatically learn from data, and is quietly infiltrating every aspect of our lives, fundamentally altering how we live. In recent years, deep learning has emerged as one of the most rapidly advancing and exciting areas within ML, with significant progress achieved through supervised, semi-supervised, and unsupervised learning. This progress has been studied in various applications in recommendation systems, web search (i.e., Google, Baidu, and Perplexity), understanding human speech, self-driving cars, and automated decision-making. Autonomous driving and advanced driver assistance systems (ADAS) (Zhao et al., 2024a) have drawn

considerable attention, especially from major automotive companies like Tesla, Google, and Uber. ADAS features, such as adaptive cruise control (ACC) (Xiao and Gao, 2010; Li et al., 2024b), lateral guidance assistance (LCA) (Zhang et al., 1990), collision avoidance system (CAS) (Zhang et al., 2024c; Aufrère et al., 2003), traffic sign recognition (Zhang et al., 2024b), and lane change assistance (Lei et al., 2024; Wang et al., 2024b), are essential components in the progression towards fully autonomous driving systems (Liu et al., 2020a; Mahdinia et al., 2020; Jiang, 2020). Early research in this field primarily focused on lane detection using mathematical models and traditional computer vision techniques, often leveraging supervised and unsupervised machine learning approaches (Kurbatova, 2020; Zhang et al., 2016, 2020, 2017). However, the research landscape has since evolved, with a growing emphasis on deep learning methodologies. Deep convolutional neural networks (DCNNs) have driven remarkable advancements in performance and become the dominant framework in both academic research and industry applications. Their superior ability to learn complex features has positioned them at the forefront of computer vision tasks. The convergence of deep learning and computer vision now serves as a crucial foundation for improving perception systems, which are integral to the development of autonomous vehicles (AVs) and advanced driver assistance systems (see Figs. 1, 6 and 12).

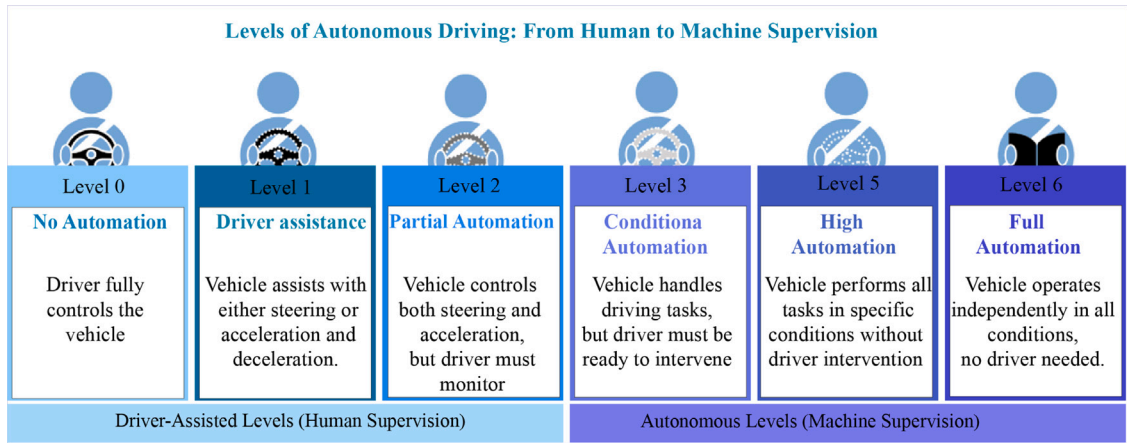


Fig. 1. Represents the levels of autonomous driving: From human to machine supervision.

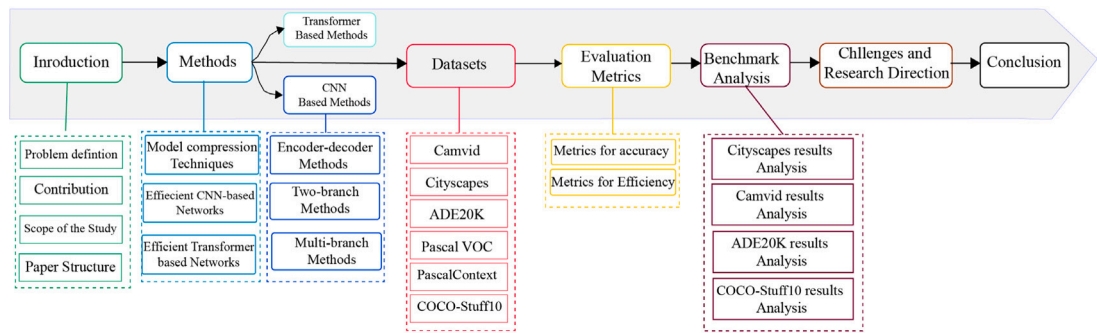


Fig. 2. The summary of the survey. Every section represented by different colors.

Early approaches of semantic segmentation were inspired by the success of CNN-based classifiers. Long et al. (Long et al., 2015) extended traditional classification neural network to perform pixel-level dense prediction through the introduction of the seminal Fully Convolutional Network (FCN). FCNs achieved remarkable performance and laid the foundation for subsequent research in semantic segmentation. Many methods have been proposed following fully convolutional network methodology (Zhong et al., 2020; Wu et al., 2019; Poudel et al., 2019). FCNs have also been successful in other tasks such as image restoration (Eigen et al., 2013), depth (Liu et al., 2015b; Eigen and Fergus, 2015), image super-resolution. Despite these advancements, the computational complexity of deep learning-based segmentation methods presents a significant bottleneck. Particularly, in applications like autonomous driving, where low-latency processing is critical. The necessity for real-time decision-making in these scenarios shows the importance of developing segmentation models that are both computationally efficient and capable of delivering precise results. In recent years, the development of efficient deep neural networks for real-time semantic segmentation has gained significant traction within the computer vision community (Li et al., 2019a; Liu and Yin, 2019; Li et al., 2020; Yang et al., 2020a). These efficient architectures are characterized by their low computational demands and reduced inference times (Sze et al., 2020), making them particularly well-suited for real-time applications across various domains. The impact of these methods has been substantial, especially in fields that demand advanced computer vision capabilities, such as autonomous vehicles (Wang et al., 2020a), robotic (Mahe et al., 2019; Chen and Pan, 2024; Alonso et al., 2020), and medical image analysis (He et al., 2023; Benabid et al., 2024; Ouahabi and Taleb-Ahmed, 2021).

Unlike traditional CNN-based methods, Transformers (Vaswani et al., 2017) have demonstrated exceptional capabilities in global context modeling and have shown strong performance on downstream tasks,

particularly when subjected to large-scale pre-training. The Vision Transformer (ViT) (Dosovitskiy et al., 2020) introduced a novel approach to image recognition by leveraging two-dimensional image patches with positional embedding as inputs. However, ViT (Dosovitskiy et al., 2020) faces a significant challenge compared to CNNs, as it requires pre-training on extensive datasets. This requirement arises because image resolution in computer vision tasks is substantially higher than in Natural Language Processing (NLP) (Chowdhary and Chowdhary, 2020), where the tokens are words. Tasks like semantic segmentation, which demand pixel-level predictions, pose additional challenges for Transformers, especially on high-resolution images, due to the quadratic relationship between the computational complexity of self-attention and image size.

The year 2021 marked a significant milestone with the introduction of Transformers to semantic segmentation (Zheng et al., 2021), in which a dense prediction is redefined as sequence to sequence task. Following this progress, Wang et al. proposed the Pyramid Vision Transformer (PVT) (Wang et al., 2021), which integrates the common feature pyramid architecture from convolutional neural networks into a Transformer-based image segmentation model. Subsequently, Xie et al. introduced SegFormer (Xie et al., 2021), a clean, efficient, and multi-scale Transformer model for image segmentation. Liu et al. contributed further with the Swin Transformer network (Liu et al., 2021a), which, upon its release, became the state-of-the-art backbone for numerous computer vision tasks, including image segmentation, surpassing traditional convolutional neural network in performance. These advancements have consistently enhanced the accuracy of image segmentation algorithms across various datasets.

1.1. Problem definition

Let $I \in \mathbb{R}^{H \times W \times 3}$ denote an input image, where H and W represent the height and width of the image, respectively. The objective of semantic segmentation is to generate a corresponding label map $S \in \mathbb{R}^{H \times W}$, where each element $S_{i,j} \in \{1, \dots, C\}$ represents the semantic class assigned to the pixel at position (i, j) . Here, C denotes the total number of predefined semantic categories (e.g., road, vehicle, pedestrian, or crack) for a perception of autonomous driving.

Over the past few years semantic segmentation has been applied to fields such as medical imaging, and remote sensing, where the emphasis is on high precision to detect subtle anatomical features such as tumors or lesions. The underlying algorithms, whether based on convolutional neural networks (CNNs), Transformers, or hybrid architectures must be tailored to these domain-specific requirements.

In the literature, most surveys and reviews on image semantic segmentation, (Hao et al., 2020; Mo et al., 2022; Minaee et al., 2021), have primarily focused on deep learning methods at a general level. Elallid et al. (2022) categorized the methods based on deep and reinforcement learning and their applications to autonomous driving. Zhu et al. (2023) reviewed methods based on weakly supervised semantic segmentation approaches. More recently, Ren et al. (2023a) has summarized few-shot learning techniques in the context of semantic segmentation. Thisanke et al. (2023) addressed the use of vision transformer in semantic segmentation. However, there has been relatively little focus on efficient deep neural networks methods specifically designed for real-time semantic segmentation (Papadeas et al., 2021; Holder and Shafique, 2022; Takos, 2020). Holder et al. (Holder and Shafique, 2022) and Takos et al. (Takos, 2020) provided overviews of real-time semantic segmentation in general computer vision. Papadeas et al. (Papadeas et al., 2021) explored the design of efficient DNNs for real-time applications within the domain of autonomous vehicles. However, these real-time semantic segmentation surveys have focused solely on CNN-based methods, and since then, there has been significant progress in convolutional neural networks. Additionally, many recent works have been published that focus on implementing pure Transformer or hybrid architectures for real-time semantic segmentation

1.2. Our contribution

We observed that there is a gap in existing literature, where current surveys lack comprehensive coverage on both Efficient CNNs and Transformer based methods or the integration of them for balancing computational complexity and inference speed in the perception systems of autonomous driving. To address this, we present a novel survey that specifically explores the application of real-time CNN and Transformer techniques within the autonomous driving domain. Unlike previous surveys on real-time semantic segmentation, this paper uniquely emphasizes the role of CNNs and Transformer models in executing the core functions of autonomous vehicles. The key contributions of this paper include:

- **Comprehensive Survey of Real-Time Semantic Segmentation:** This paper provides an exhaustive survey of the state-of-the-art real-time semantic segmentation techniques specifically tailored for autonomous vehicles, covering both CNN-based methods and emerging Transformer-based approaches.
- **Analysis of Efficiency-Accuracy Trade-offs:** We analyze the trade-offs between accuracy and inference speed across various real-time semantic segmentation models, highlighting the strengths and weaknesses of different architectural designs, such as two-branch networks, lightweight models, and hybrid frameworks.

- **Identification of Key Challenges:** We identify and discuss the major challenges in the field, such as the integration of Transformer architectures into real-time segmentation, the dependency on high-quality training data, the domain gap across datasets, and the application of these models on edge devices.
- **Proposal of Future Research Directions:** The paper proposes several future research directions, including the development of scalable and robust segmentation models, the exploration of energy-efficient inference, the potential of multi-modal segmentation, and the importance of addressing ethical and bias considerations in real-time segmentation.
- **Discussion of Emerging Trends:** We highlight emerging trends in the field, such as the application of lightweight Transformer designs and knowledge distillation techniques from large visual models, suggesting how these innovations could shape the next generation of real-time semantic segmentation systems.
- **Resource Compilation for Researchers:** The paper compiles an extensive list of datasets, evaluation metrics, and state-of-the-art models, serving as a valuable resource for researchers and practitioners looking to advance the field of real-time semantic segmentation.
- **Quantitative Evaluation and Comparative Study:** We provide a detailed quantitative evaluation of real-time semantic segmentation models on standardized hardware, offering an intuitive comparison of the speed-accuracy trade-offs and helping researchers make informed decisions about model selection.

1.3. Scope of the study

This survey provides a comprehensive overview of the evolution of real-time semantic segmentation, covering key methods, datasets, applications, and challenges. We emphasize the importance and complexity of achieving high-accuracy results in real-time, particularly on computationally constrained devices. The survey explores real-time semantic segmentation techniques, focusing on both convolutional neural networks and Transformer-based approaches. Given the vast number of pre-prints and published works in this domain, our survey highlights the most seminal and impactful contributions.

1.4. Paper structure

The structure of this survey is organized as follows: Overall, Fig. 2 shows the pipeline of our survey. In Section 2 we present a detailed overview of state-of-the-art techniques for real-time semantic segmentation in autonomous driving. Section 2.1 explores into network compression strategies aimed at optimizing computational efficiency. We then categorize and discuss CNN-based segmentation methods in Section 3.1, followed by an in-depth exploration of Transformer-based architectures in Section 3.2. Section 4 examines the datasets and evaluation metrics commonly used in this domain, providing a foundation for our benchmark analysis in Section 5. Lastly, we highlight key challenges and future research directions in Section 6.1 before concluding in Section 7. The acronyms are summarized in Table 8.

2. Methods: A survey

Perception of the environment is a critical challenge for intelligent robots and autonomous driving systems, where precise and real-time scene understanding is paramount. Recent advances in deep Convolutional Neural Networks (CNNs) have driven significant progress in semantic segmentation, enabling pixel-level classification that enhances the comprehension of complex driving scenarios (Wen and Jo, 2022; Han et al., 2023). Beyond the recognition of common objects and obstacles, the ability to detect unexpected road hazards, such as debris and cracks (Zhang et al., 2024a; Wang et al., 2024d; Zhang et al., 2024d), is essential for the safe and reliable operation of autonomous

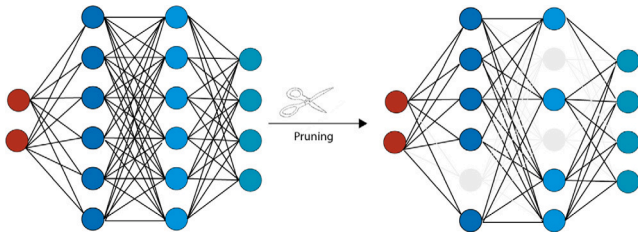


Fig. 3. A simplified version of neuron and weight pruning in a neural network.

vehicles, highlighting the need for robust, adaptive perception systems. This section provides a comprehensive overview of the lightweight and efficient model designs that have been widely adopted and extensively studied in semantic segmentation tasks.

2.1. Model compression techniques

Model compression involves reducing the size of a large neural network to enhance real-time performance or minimize parameter count without significantly compromising accuracy. Common model compression techniques include network pruning, neural architecture search (NAS), knowledge distillation, and quantization.

2.1.1. Network pruning

Pruning eliminates unnecessary parameters from a network. The typical steps involve training a large network, assessing the importance of each parameter, removing less critical parameters, and fine-tuning the pruned network to recover some of the accuracy lost during pruning. This approach leverages the ease of training large models to high accuracy and derives a more compact model with minimal accuracy loss. Pruning techniques have been utilized across various levels of sparsity, ranging from fine-grained approaches (Zhang et al., 2018b; Yu et al., 2018a), which involve the removal of individual weights or neurons from a model, as illustrated in Fig. 3, to coarse-grained approaches (Wen et al., 2016; He et al., 2017; Chen et al., 2020; Jiang et al., 2022), which target the elimination of entire filters or channels.

2.1.2. Neural architecture search (NAS)

Neural architecture search employs reinforcement learning to concurrently optimize both the model architecture and its parameters (Dou et al., 2023; Wu et al., 2022b; Zhu et al., 2024a). Essentially, it identifies the optimal network architecture within a defined search space using specific search strategies. By incorporating metrics such as accuracy and inference latency, NAS can yield network structures that are highly competitive in lightweight applications.

2.1.3. Knowledge distillation

Knowledge distillation transfers the knowledge encapsulated in a large teacher model to a smaller student model (Hinton et al., 2015; Chen et al., 2021; Liu et al., 2019, 2022b; Wu et al., 2021a; Kothandaraman et al., 2021). This process involves guiding the student model to mimic the output of the teacher model, thereby enhancing the student model's accuracy and producing a compact model with improved performance. Fig. 4 illustrate a general framework for knowledge distillation technique. For instance MobileSAM (Zhang et al., 2023a,b) distilled the foundational vision model SAM (Kirillov et al., 2023), which has a strong zero-shot performance into more compact network with fast inference while performing on par with its teacher network.

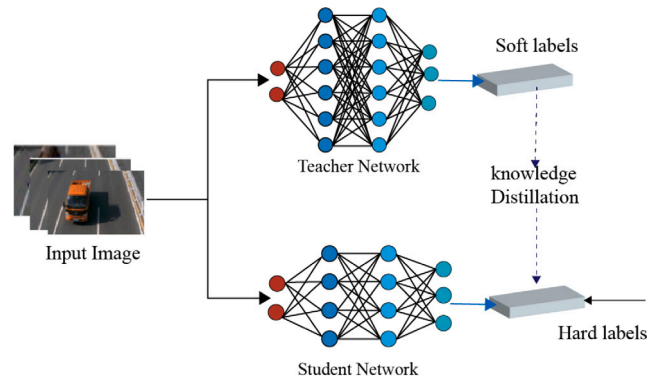


Fig. 4. A simplified version of knowledge distillation technique.

2.1.4. Quantization

Model quantization reduces the numerical precision of a model's parameters to achieve compression and accelerate inference speed. This technique is frequently employed in model deployment, particularly in resource-constrained environments. In quantization the model usually rendered with low-bit precision operands (Hubara et al., 2018; Mishra et al., 2017), using linear quantization methods (Fang et al., 2020), or binary methods (Razani et al., 2021; Liu and Liu, 2023)

2.2. Efficient CNN-based networks

In the previous section, we discussed the importance of optimizing Convolutional Neural Networks (CNNs) for low-power applications, often requiring techniques such as weight pruning, quantization, and model compression. While these methods position CNNs as major candidates for edge devices operating under stringent energy constraints, they frequently come at the cost of reduced accuracy. Despite this trade-off, the field of computer vision has witnessed significant advancements since the seminal introduction of AlexNet (Krizhevsky et al., 2012), which set a new benchmark in performance. Through architectural innovations and methodical optimizations, the landscape of CNNs has evolved, leading to state-of-the-art models that have repeatedly surpassed previous performance records while concurrently reducing computational demands. The advent of ResNet (He et al., 2016a,b), which introduced residual connections, marked a pivotal shift in the design of deep learning architectures. This breakthrough facilitated the training of substantially deeper models by providing an alternative gradient pathway during back-propagation, effectively addressing the vanishing gradient problem, as depicted in Fig. 5. Consequently, these deeper architectures have achieved unprecedented performance across a broad range of benchmark tasks. ResNet's influence has been transformative, serving as a foundational architecture that has inspired a multitude of contemporary models. However, as high-performing CNNs, characterized by large numbers of parameters and Floating Point Operations Per Second (FLOPS), continued to achieve remarkable results, the emergence of lightweight CNN architectures has proven to be a compelling alternative. Models such as EfficientNet (Tan and Le, 2019, 2021), MobileNet (Howard et al., 2017; Sandler et al., 2018; Howard et al., 2019), ShuffleNet (Zhang et al., 2018c; Ma et al., 2018), SqueezeNet (Iandola et al., 2016), and ESPNet (Mehta et al., 2018, 2019) have made significant steps in optimizing CNNs by introducing excessively deep and densely interconnected structures. These architectures align with the core principles established by ResNet and VGG (Simonyan, 2014), emphasizing a careful balance between model complexity, parameters, and computational efficiency.

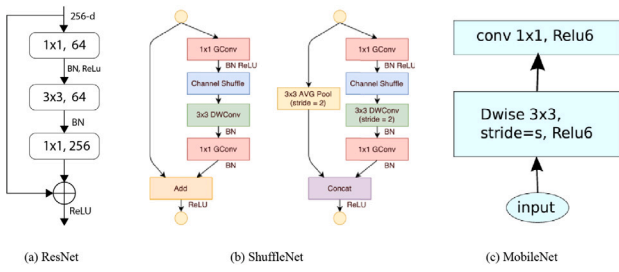


Fig. 5. Important classification networks. From left, Residual block (He et al., 2016a), ShuffleNet (Zhang et al., 2018c), and MobileNet (Howard et al., 2017).

2.3. Efficient transformer-based networks

The Vision Transformer (ViT) (Dosovitskiy et al., 2020) revolutionized the field by adapting the transformer architecture to large-scale image recognition tasks, achieving state-of-the-art performance that surpasses traditional CNNs (Dosovitskiy et al., 2020). Building upon the success of ViTs, subsequent research has aimed to integrate spatial inductive biases to improve their performance and stability (Dai et al., 2021; Guo et al., 2022), develop more efficient self-attention mechanisms (Dong et al., 2022), and extend the application of ViTs to a broader array of computer vision tasks (Esser et al., 2021; Zhang et al., 2022). Despite their superior performance over CNNs across various vision tasks, ViTs typically demand significant computational resources and memory, making them less suitable for deployment on mobile devices with constrained resources (Liu et al., 2021a; Touvron et al., 2021). This challenge has spurred researchers to explore techniques for making ViTs more lightweight and compatible with mobile platforms (Mehta and Rastegari, 2022; Vasu et al., 2023a). For instance, MobileViT (Mehta and Rastegari, 2021) employs a hybrid architecture that combines lightweight MobileNet blocks with multi-head self-attention (MHSA) modules. Similarly, EfficientFormer (Li et al., 2022b) introduces a dimension-consistent design paradigm to push the boundaries of latency-performance trade-offs. These innovations in lightweight ViTs have set new benchmarks in balancing performance and latency on mobile devices, often outperforming previous lightweight CNNs (Vasu et al., 2023b; Sandler et al., 2018). The success of these lightweight ViTs is largely attributed to their multi-head self-attention modules, which excel at learning global representations.

The success of varying lightweight CNNs and Transformer architecture design highlight their ability to deliver competitive performance while maintaining efficiency, making them particularly well-suited for deployment in resource-constrained environments. CNNs continue to play a pivotal role across various domains, including classification, object detection, and segmentation. The research communities in object detection and segmentation are increasingly converging, recognizing the critical role that classification CNNs play as the backbone for these tasks alongside the Transformer and the emerging state space models such as vision Mamba (Zhu et al., 2024b). Therefore, in the subsequent sections, we will elaborate on different network architectures to give broader perspective on enhancing EdgeAI capabilities

3. Methods categorization

3.1. Semantic segmentation based CNN networks

In semantic segmentation, the main objective is to assign a categorical label to every pixel in an image, which plays a significant role in road scene understanding. The success and advances in deep convolutional neural networks (CNN) models (Szegedy et al., 2015; He et al., 2016a; Krizhevsky et al., 2012; Simonyan, 2014) have a remarkable impact on pixel-wise semantic segmentation progress due

to the rich hierarchical features (Ronneberger et al., 2015; Zheng et al., 2015; Long et al., 2015; Liu et al., 2015a; Lin et al., 2016). Usually, to obtain a more delicate result from such deep network, it is essential to retain high-level semantic information when using low-level details. However, training such a deep neural network requires a large amount of data, but in many practical cases, only a limited number of training examples are available. One way to overcome this problem is by employing transfer learning through a network that pre-trained on a big dataset then fine-tune on the targeted dataset, as done in Long et al. (2015), Zhou et al. (2018). Another solution for such a problem is by performing extensive data augmentation, as done in UNet (Ronneberger et al., 2015). In addition to data augmentation, the model architecture should be designed to allow the propagation of the information from low levels to the corresponding high levels in a much easier way such as U-Net.

3.1.1. Encoder-decoder based networks

Encoder-decoder based models have been successfully used in semantic segmentation to overcome some of limitation in the original FCN as mentioned in Section 1. As shown in Fig. 7, in this approach the encoder constructed to reduce the spatial dimension of the input image using a pooling layer, while the decoder uses deconvolutional layer or upsampling to gradually restore the lost spatial dimension. Noh et al. (2015) proposed a semantic segmentation model based on transposed convolution, the model utilizes VGG-16 as encoder while the decoder composed of transposed convolution and unpooling layers. Badrinarayanan et al. (2017) proposed an important network known as SegNet. It is based on encoder-decoder architecture, and similar to Noh et al. (2015) network, but uses 13 convolutional layers in the encoder. SegNet main contribution was using the pooling indices computed in the encoder through the max-pooling step to perform the upsampling in the corresponding decoder.

Another important model in this category U-shape structure methods such as U-Net (Ronneberger et al., 2015), which uses the skip-connections to acquire the multi-level features and help restore the lost information in the encoding part. U-Net initial proposed for medical /biomedical segmentation, which inspired by the encoder-decoder and FCN models. Since then U-Net has inspired a lot of models in medical and non-medical fields. Figure illustrates the original U-Net architecture.

3.1.2. Two-branch networks

In this section, we review the latest advancements in real-time semantic segmentation models, specifically focusing on their trade-off accuracy/inference time on scenes understanding benchmarks. The models are categorized based on the strategies employed to achieve efficiency, with particular attention to two-branch network architectures, which are prominent among state-of-the-art approaches in real-time semantic segmentation (Wu et al., 2017; Poudel et al., 2018; Yu et al., 2018b, 2021). The SS model (Wu et al., 2017) exemplifies a multi-branch network design, utilizing two branches: one for the input image and another for a half-resolution version. SS also introduces spatial sparsity into this architecture, significantly reducing computational cost by a factor of 25. This efficiency is further enhanced by incorporating in-column and cross-column connections and eliminating residual units. ContextNet (Poudel et al., 2018), another two-branch model, implements a low-resolution branch for cost-effective and accurate segmentation, while a high-resolution sub-network refines the segmentation details. ContextNet also employs depth-wise separable convolutions to accelerate inference and integrates bottleneck residual blocks. Key to its performance is network compression and the use of pyramid representations, enabling real-time segmentation with minimal memory overhead. Additionally, ContextNet applies pruning techniques to reduce network parameters. Fast-SCNN (Poudel et al., 2019) introduces a “learning to downsample” module that computes

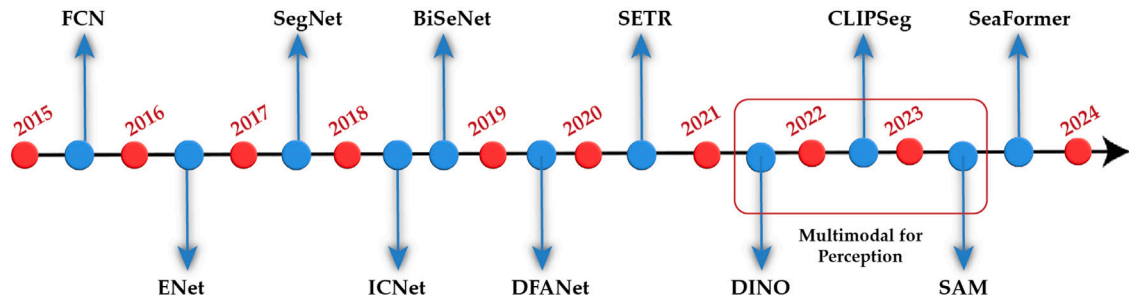


Fig. 6. Timeline of significant advancements in real-time semantic segmentation for autonomous vehicles, highlighting key milestones and the evolution of techniques from early lightweight networks to recent Transformer-based models.

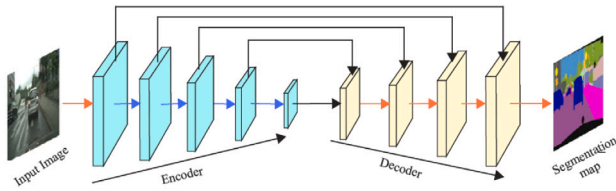


Fig. 7. Schematic of the Encoder-Decoder model.

low-level features for multiple resolution branches in parallel. It also includes a global feature extractor to capture the global context necessary for semantic segmentation. Fast-SCNN leverages depth-wise separable convolutions and residual bottleneck blocks [26] to enhance speed while minimizing parameters and computational cost. The Bilateral Segmentation Network (BiSeNet) (Yu et al., 2018b) introduces a feature fusion module to efficiently combine features, along with an attention refinement module to filter features at each stage. This design improves precision without resorting to upsampling operations, thereby maintaining low computational costs. BiSeNet V2 (Yu et al., 2021), an evolution of BiSeNet, achieves a strong balance between speed and accuracy by featuring a Guided Aggregation Layer for fusing features from the Detail and Semantic branches, and a booster training strategy. The Semantic Branch employs fast-downsampling to enhance feature representation and rapidly increase the receptive field. Similarly, DSANet (Elhassan et al., 2021b) effectively balances fine-grained detail and high-level semantic information by employing a dual-branch architecture, where a shallow spatial detail branch is paired with a deep semantic context branch. This design ensures the preservation of intricate spatial features while capturing essential contextual information. CABiNet (Kumaar et al., 2021) incorporates a lightweight global information aggregation module within the Semantic Branch, effectively capturing the contextual relevance needed for accurate segmentation. On the other hand, STDC (Fan et al., 2021) introduces the Laplacian operator and calculates edge loss during training, guiding the network's learning process to enhance the capture of detailed features. A representative structures of these networks is presented in 8 and Table 1 includes more methods that belong to the same category.

3.1.3. Multi-branch networks

In order to extract feature information of different scales of the input image, ICNet (Zhao et al., 2018) downsample the input image to three different sizes of $1/2$, $1/4$ and $1/8$, and then processes these images through separate branches respectively. For images with lower resolution, the pre-trained ResNet is used as a feature extractor to extract high-level features; for images with higher resolution, three consecutive convolutions are used to extract the necessary edge information. To address the challenge of small target segmentation in low-resolution images, ICNet integrates features from $1/16$, $1/8$, and

$1/4$ of the original image size with convolutional features. This multi-branch approach effectively balances high-, medium-, and low-level feature retention while mitigating the computational demands typically associated with large feature maps. However, ICNet's method of reducing feature maps by cropping leads to a loss of detail, particularly in boundary information, which can negatively impact segmentation accuracy. DFANet (Li et al., 2019a) addresses these limitations by leveraging an enhanced lightweight backbone network, Xception (Chollet, 2017), and introduces a design with three sub-networks. The outputs from the initial and final stages of each sub-network are cascaded and serve as the input for the subsequent sub-network, thereby preserving and refining the features extracted at earlier stages.

3.2. Semantic segmentation based transformer networks

The Vision Transformer (ViT) (Dosovitskiy et al., 2020) introduced a paradigm shift in image recognition by reframing images as sequences of patches, processed through a deep stack of transformer layers. This framework was soon extended by models such as DeiT (Touvron et al., 2021), Fact (Jie and Deng, 2023), CrossFormer (Wang et al., 2023), and DViT (Yao et al., 2023b), which have set new benchmarks across various image processing tasks. In semantic segmentation, SETR (Zheng et al., 2021) adapts this pure transformer architecture for dense prediction, combining a transformer encoder with various CNN-based decoders to achieve leading performance. The Swin Transformer (Liu et al., 2021a) mitigates computational inefficiencies by reducing redundant operations, thereby lowering computational costs. However, despite these advancements, transformer-based models still require vast amounts of training data to approach the performance levels of CNNs, particularly in tasks demanding fine-grained, dense annotations.

Transformers like those in Lee et al. (2022), Li et al. (2022a) have recognized the necessity of hierarchical processing for dense prediction, incorporating multi-scale structures and pyramid modules to enhance their effectiveness. While transformers excel at capturing global, long-range dependencies, they often lack the ability to capture local connections and translation invariance—key strengths of CNNs. This has led to hybrid approaches SegTransConv (Fan et al., 2023) that attempt to fuse the complementary strengths of CNNs and transformers. Despite these efforts, achieving the delicate balance between real-time inference and low-latency processing remains a significant challenge.

To address this, recent research has explored lightweight architectures that integrate transformers more efficiently. LET Net (Xu et al., 2023a), for example, positions the transformer within a capsule network, while TopFormer (Zhang et al., 2022) and P2AT (Elhassan et al., 2024) utilizes transformers as auxiliary components in the decoder to improve boundary refinement. Yet, a robust solution for harmonizing global and local information within the constraints of real-time processing remains an open research question, and further exploration is required to fully unlock the potential of these hybrid models (see Table 2).

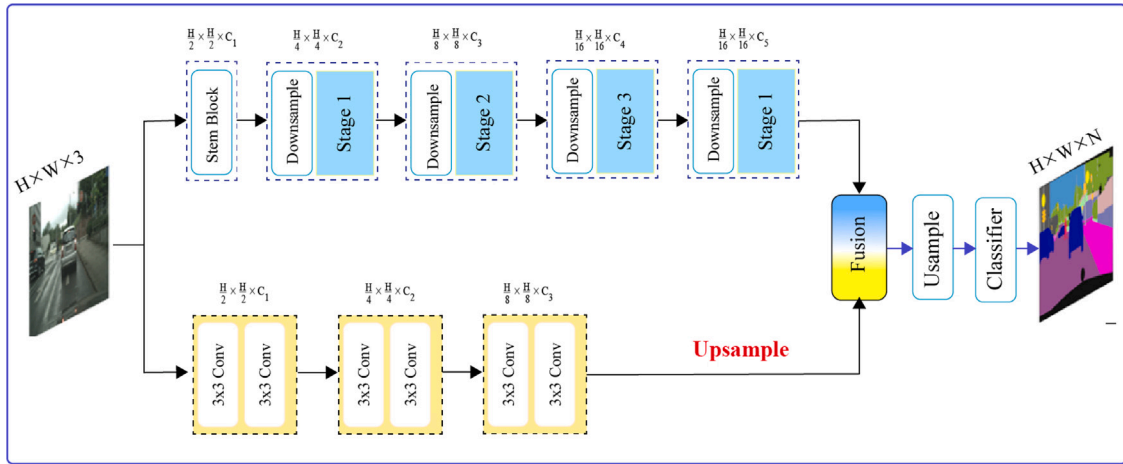


Fig. 8. Schematic of the dual-branch network for efficient real-time semantic segmentation.

Table 1
Popular common semantic segmentation architectures.

Method category	Representative references
Two-branch Networks (Section 3.1.2)	BiSeNet (Yu et al., 2018b), DSANet (Elhassan et al., 2021b), BiSeNetV2 (Yu et al., 2021) STDC (Fan et al., 2021) DDRNet (Hong et al., 2021), LBARNet (Hu and Zhou, 2023) BiSeNetV3 (Tsai and Tseng, 2023), BFMNet (Liu et al., 2023), STDBNet (Ren et al., 2023b) Fast-SCNN (Poudel et al., 2019), CABiNet (Kumaar et al., 2021), ContextNet (Poudel et al., 2018) MAFNet (Zhao et al., 2024b), BiSeNetFormer (Rosi et al., 2024), MLFNet (Fan et al., 2022), BiConvNet (Wu and Zhu, 2024), LBN-AA (Dong et al., 2020), STRNet (Wang et al., 2024c) BiAttnNet (Li et al., 2021a), LDANet (Zhu et al., 2024c), MCFNet (Fang et al., 2023) SFRSeg (Singha et al., 2023), BiDENet (Ye et al., 2024), BiDGANet (Liao et al., 2023) CIDNet (Dong et al., 2023b), BMSeNet (Zhao et al., 2024c)
Multi-branch Networks (Section 3.1.3)	ICNet (Zhao et al., 2018), DFANet (Li et al., 2019a), LPS-Net (Zhang et al., 2023c) Optimized HRNet (Wu et al., 2021b), Lite-HRNet (Yu et al., 2021b), DMA-Net (Weng et al., 2022) TBFormer (Wei and Wei, 2024), PIDNet (Xu et al., 2023b), LCFNet (Yang et al., 2024)
Encoder-decoder Networks (Section 3.1.1)	AGLNet (Zhou et al., 2020), FBRNet (Qu et al., 2024), ELANet (Yi et al., 2023) LMANet (Hu and Liu, 2023), SPFNet (Elhassan et al., 2022a), AFPNet (Hyun et al., 2021) SEFANet (Wang et al., 2024a), EAAMNet (Su et al., 2024), LRDNet (Zhuang et al., 2021) , MFAFNet (Lu et al., 2023), ZMNet (Li et al., 2024a), SFANet (Weng et al., 2021) BADNet (Peng et al., 2021), PPANet (Elhassan et al., 2021a), ENet (Paszke et al., 2016) FCN (Long et al., 2015), LSNet (Sheng et al., 2022), DABNet (Li et al., 2019b) WFDCNet (Hao et al., 2021), ANet-16 (Song et al., 2024), S ² -FPN (Elhassan et al., 2022b) CSP (Zhou et al., 2023), EBUNet (Shen et al., 2023), RELAXNet (Liu et al., 2022a) LCNet (Shi et al., 2024), LMFFNet (Shi et al., 2024), FNANet (Lan et al., 2024)
Transformer Networks (Section 3.2)	DTMC-Net (Hu et al., 2024), SegTransConv (Fan et al., 2023), HAFormer (Xu et al., 2024) SeaFormer-S (Wan et al., 2023), BiSeNetFormer (Rosi et al., 2024), LETNet (Xu et al., 2023a) TopFormer (Zhang et al., 2022), RTFormer (Wang et al., 2022), AFFormer (Dong et al., 2023a) SSFormer (Shi et al., 2022), TBFormer (Wei and Wei, 2024), LCBFormer-Net (Dong et al., 2020)

Table 2
Methods for real-time semantic segmentation.

Method	Year	backbone	Highlight
ENet	Paszke et al. (2016)	2015	• Efficient Neural Network With low latency operation
SQ	Treml et al. (2016)	2016	• Combines a SqueezeNet-like encoder, ELU activation functions, parallel dilated convolutions, and a decoder with SharpMask-like refinement modules
ERFNet	Romera et al. (2017)	2017	• Lightweight Network with factorized Convolution
ContextNet	Poudel et al. (2018)	2018	• Context information Aggregation from multi-level resolution
ESPNet	Mehta et al. (2018)	2018	• Efficient spatial pyramid that utilize dilated convolution
ESPNetV2	Mehta et al. (2019)	2018	• Efficient spatial pyramid that utilize dilated convolution
ICNet	Zhao et al. (2018)	2018	• Effective extraction of information at multiple scales, and the number of branches • can be adjusted according to the image size.
BiSeNet	Yu et al. (2018b)	2018	• Add detail branches to retain spatial details of the input image and improve the segmentation accuracy of the target boundary • The detail branch could reduce the inference speed
CGNet	Wu et al. (2018)	2018	• Enhances joint features by integrating local and global contextual information. • Ensures contextual information is captured across all stages of the network
GUN	Mazzini (2018)	2018	• GUM enriches traditional upsampling by adding a learnable transformation for semantic maps.
ShelfNet	Zhuang et al. (2019)	2019	Introduces a novel multi-encoder-decoder architecture resembling a shelf, enhancing segmentation accuracy by leveraging multiple deep and shallow paths. • Proposes a shared-weight strategy in residual blocks to reduce parameters without compromising performance
DABNet	Li et al. (2019b)	2019	• combining depth-wise asymmetric and dilated convolutions using depth-wise asymmetric bottleneck . • Expands the receptive field while maintaining minimal computational overhead.
DFANet	Li et al. (2019a)	2019	• Maximize multi-scale feature aggregation with less computation cost
Fast-SCNN	Poudel et al. (2019)	2019	• implements a novel module for simultaneous computation of low-level features across multiple resolution branches.
LiteSeg	Emara et al. (2019)	2019	• Utilizes depth-wise separable convolution to enhance model speed and efficiency. • Explores a deeper version of ASPP with short and long residual connections
ThunderNet	Xiang et al. (2019)	2019	• Introduces turbo unified network, a lightweight and efficient model that . utilizes a minimal backbone derived from ResNet18 for enhanced speed • It combines pyramid pooling with a customized decoder for optimized performance.
LBN-AA	Dong et al. (2020)	2020	• Distinctive Atrous Spatial Pyramid Pooling
AGLNet	Zhou et al. (2020)	2020	• Explore hierarchical feature using factorized attention module
NDNet45	Yang et al. (2020b)	2020	• Strategy to augment small object of road scene
DSANet	Elhassan et al. (2021b)	2021	• Integrates detail branches to retain spatial details • Uses dilated spatial attention to refine semantic feature in the early stages
Re-BiSeNet	Fan et al. (2021)	2021	• Enhances spatial information learning within the low-level layers with detail aggregation module
BiSeNetV2	Yu et al. (2021)	2021	• Uses shallow layers and wide channels to capture low-level details • Uses semantics branch to obtain high-level semantic context
DDRNet	Pan et al. (2022)	2022	• Leverages dual deep branches and bilateral fusions to produce superior detail quality compared to existing two-pathway methods.
TopFormer	Zhang et al. (2022)	2022	• Utilizes multi-scale token inputs to enhance semantic feature representation, improving segmentation accuracy
AutoSegEdge	Dou et al. (2023)	2023	• Employs FLOPs and latency as search constraints to translating NAS into a Multi-Task Learning framework. • Automatic method to learn the weights of sub-objective loss functions, enhancing stability. • Knowledge distillation for improved accuracy
SegTransConv	Fan et al. (2023)	2023	• Combines transformer layers with CNN modules to improve context aggregation and feature extraction. • Utilizes STDCNet as a teacher network to enhance model performance through knowledge distillation.
STDBNet	Ren et al. (2023b)	2023	• Merges branch features using a redesigned spatial attention mechanism for enhanced feature augmentation
CSFCN	Li et al. (2023a)	2023	• Employs learnable sampling to calibrate spatial features by splitting features into sub-groups.
MAFNet	Zhao et al. (2024b)	2024	• Enhances boundary extraction and fine-grained classification • Introduces cross-fusion to balance and effectively merge feature information between the dual branches for optimized performance

(continued on next page)

Table 2 (continued).

HAFormer	Xu et al. (2024)	2024	- Efficient Transformer module that reduces the computational complexity. - Enhances predictive accuracy by merging diverse feature representations using correlation-weighted fusion
TBFormer	Wei and Wei (2024)	2024	- Context branch leverages transformers for rich global context and refines multi-scale information - Spatial branch utilizes small stride convolutions to maintain spatial details. - Edge branch enhances boundary features using Sobel operator integration.

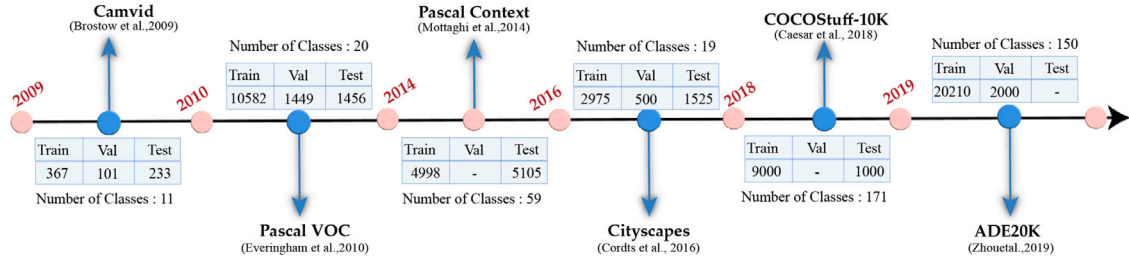


Fig. 9. Timeline of significant advancements in real-time semantic segmentation for autonomous vehicles, highlighting key milestones and the evolution of techniques from early lightweight networks to recent Transformer-based models.

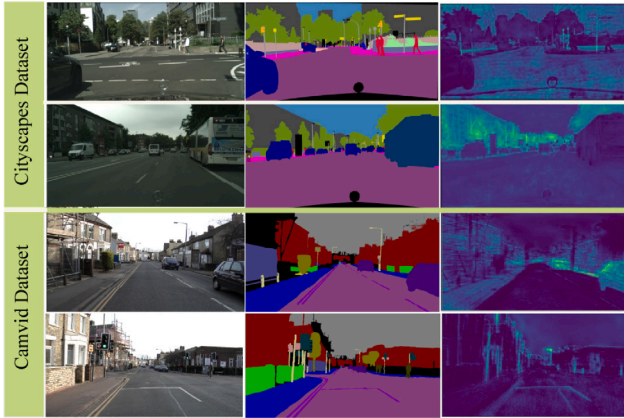


Fig. 10. A sample of Cityscapes and Camvid datasets. From left: Original image, ground truth, and feature visualization.

4. Datasets and metrics for semantic segmentation

This section offers a comprehensive overview of the most widely used datasets for training and evaluating deep learning models in real-time semantic segmentation. These datasets were chosen based on their ability to capture diverse environmental conditions. Fig. 10 provides a summary of the key datasets frequently used in semantic segmentation tasks.

4.1. Datasets

We first present a detailed examination of several widely recognized datasets for semantic segmentation as represented in Fig. 9, followed by an in-depth discussion of the standard metrics used to evaluate the performance of segmentation models.

4.1.1. ADE20K dataset

ADE20K (Zhou et al., 2019) is a benchmark dataset for scene parsing, comprising 20,210 training images and 2000 validation images. The dataset poses significant challenges due to its inclusion of 150 diverse object categories, ranging from specific objects like dogs and cars to broader categories such as roads and sky. The test set is

currently not publicly available, adding an extra layer of complexity for benchmark model performance.¹

4.1.2. Cityscapes dataset

The Cityscapes dataset (Cordts et al., 2016), a very important in semantic segmentation research, encompasses a comprehensive collection of urban street scenes from 50 European cities. The dataset is carefully curated, comprising 20,000 images with coarse annotations and 5000 images with fine annotations. Typically, the coarse-annotated subset is leveraged during the pre-training phase to enhance the model's generalization capabilities. The fine-annotated set is partitioned into 2975 images for training, 500 for validation, and 1525 for testing. Importantly, the test set's ground truth annotations remain undisclosed, with evaluation facilitated through an online server.² A distinguishing feature of this dataset is its high-resolution imagery, captured at 2048×1024 pixels, offering a rich and detailed input for training state-of-the-art segmentation models.

4.1.3. Camvid dataset

The CamVid dataset (Brostow et al., 2009) is a widely recognized benchmark in semantic segmentation, offering annotated driving scene frames extracted from five distinct video sequences. It includes 701 frames that have been methodically labeled across 32 semantic classes. In subsequent work by Sturges et al. (2009), these frames were systematically divided into training, validation, and testing subsets, comprising 367, 100, and 233 images, respectively. Notably, the dataset has been further refined to focus on 11 primary semantic classes, with the remaining 21 classes consolidated into a 'void' category for model training and evaluation. This dataset has been instrumental in advancing research on real-world scene understanding.³

4.1.4. PASCAL VOC 2012 dataset

PASCAL VOC 2012 (Everingham et al., 2010) serves as a seminal dataset in the field of object recognition and segmentation. It features 1464 images for training, 1449 for validation, and 1456 for testing, with annotations covering 20 distinct object categories and a background class. Unlike some other datasets, PASCAL VOC 2012 allows

¹ <https://groups.csail.mit.edu/vision/datasets/ADE20K/>

² <https://www.cityscapes-dataset.com/dataset-overview/>

³ <https://mi.eng.cam.ac.uk/research/projects/VideoRec/CamVid/>

public access to the ground truth of its test set, facilitating transparent and consistent performance evaluation. Moreover, the dataset's augmented version (Hariharan et al., 2011), which expands the training set to 10,582 images, is frequently employed to boost model robustness and generalization.

4.1.5. PASCAL context dataset

The PASCAL Context dataset (Mottaghi et al., 2014) builds upon the PASCAL VOC 2010 segmentation challenge, enriching the annotations to encompass entire scene representations. This dataset includes 10,103 images allocated for training and validation, alongside 9637 images designated for testing. Although it supports over 400 semantic classes, practical usage typically focuses on the 59 most commonly occurring classes, given the sparse distribution of labels. PASCAL Context has become a crucial resource for developing models capable of comprehensive scene understanding and segmentation in complex environments.

4.1.6. COCOStuff-10K dataset

The COCO-Stuff dataset, introduced by Caesar et al. (2018), represents an important benchmark for scene understanding in computer vision. Specifically, the COCO-Stuff-10K subset comprises 10,000 densely annotated images from the broader COCO-Stuff dataset, with 9000 images reserved for training and 1000 for testing. This dataset includes a total of 182 categories, encompassing 91 "thing" classes and 91 "stuff" classes. However, due to the exclusion of annotations for 11 "thing" classes, the dataset effectively includes 171 annotated categories. The diversity and complexity of the scenes and categories in COCO-Stuff-10K make it a particularly challenging benchmark for real-time semantic segmentation tasks.

4.2. Evaluation metrics

In this section, we discuss the evaluation metrics for semantic segmentation, focusing on two critical aspects: accuracy and efficiency. We introduce the metrics commonly used for each of these perspectives.

4.2.1. Metrics for accuracy

For the evaluation of accuracy, we consider a scenario with $k + 1$ categories, where k represents the number of classes, and one additional category represents the background. Let P denote the total number of pixels. For instance, P_{ij} refers to the count of pixels that are predicted as belonging to the j th class but actually belong to the i th class. Therefore, P_{ii} , P_{ij} , and P_{ji} correspond to the True Positive (TP), False Positive (FP), and False Negative (FN) counts, respectively. Below, we detail five metrics commonly used to assess the accuracy of semantic segmentation: Pixel Accuracy (PA), Mean Pixel Accuracy (MPA), Intersection over Union (IoU), Mean Intersection over Union (mIoU), and Frequency-Weighted Intersection over Union (FWIoU).

Pixel Accuracy (PA): Pixel Accuracy measures the proportion of correctly classified pixels relative to the total number of pixels. Formally, PA is defined as the ratio of True Positives to the sum of True Positives and False Negatives, expressed as $\frac{TP}{TP+FN}$. This metric is computed by summing the diagonal elements of the confusion matrix, which represent the true positive counts for each class, and then dividing by the total number of pixels.

$$PA = \frac{\sum_{i=0}^k P_{ii}}{\sum_{i=0}^k \sum_{j=0}^k P_{ij}} \quad (1)$$

Where P_{ii} and P_{ij} denote the number of true positives and false positives, respectively.

Mean Pixel Accuracy (MPA): Mean Pixel Accuracy extends the concept of Pixel Accuracy by computing the accuracy on a per-class basis and then averaging these values across all classes. Instead of measuring the overall accuracy, MPA provides a more granular view by evaluating the accuracy for each class individually. Specifically, it

calculates the ratio of true positives to the total number of samples for each class (i.e., the sum of elements in the corresponding row of the confusion matrix). The final MPA is obtained by averaging these per-class accuracies, offering a balanced measure that accounts for class imbalances.

$$MPA = \frac{1}{k+1} \sum_{i=0}^k \frac{P_{ii}}{\sum_{j=0}^k P_{ij}} \quad (2)$$

Intersection over Union (IoU): Intersection over Union (IoU) is a crucial metric for evaluating the accuracy of image segmentation tasks. It measures the overlap between the predicted segmentation mask and the ground truth mask by calculating the ratio of correctly predicted pixels (intersection) to the total number of pixels in either the predicted or ground truth mask (union). Specifically, IoU is computed as the number of pixels shared by both the predicted and actual regions divided by the number of pixels that exist in either region, effectively penalizing both false positives and false negatives. This metric is particularly valuable for tasks where precise delineation of boundaries is essential, as it provides a more stringent measure of prediction accuracy compared to simpler metrics like pixel accuracy.

$$IoU = \frac{\sum_{i=0}^k P_{ii}}{\sum_{i=0}^k \sum_{j=0}^k P_{ij} + \sum_{i=0}^k P_{ii} - \sum_{i=0}^k P_{ii}} \quad (3)$$

Mean Intersection over Union (mIoU): The Mean Intersection over Union metric represents the average IoU across all classes. It is computed by taking the mean of the IoU values for each class, ensuring that the performance is evenly evaluated across all categories, regardless of their individual sizes. Formally, it is defined as:

$$mIoU = \frac{1}{k+1} \sum_{i=0}^k \frac{P_{ii}}{\sum_{j=0}^k P_{ij} + \sum_{j=0}^k P_{ji} - P_{ii}} \quad (4)$$

Where P_{ii} refers to the number of true positives, while P_{ji} and P_{ij} represent false negative and false positives, respectively.

Frequency-Weighted Intersection over Union (FWIoU): Frequency-Weighted Intersection over Union is an enhancement of the mIoU metric that accounts for the frequency of occurrence of each class in the dataset. This weighting mechanism adjusts the contribution of each class to the overall metric, giving more importance to classes that appear more frequently. The FWIoU is particularly useful in scenarios where the dataset contains a significant imbalance among class frequencies, as it provides a more realistic measure of a model's performance in such cases.

$$FWIoU = \frac{1}{\sum_{i=0}^k \sum_{j=0}^k P_{ij}} \sum_{i=0}^k \frac{\sum_{j=0}^k P_{ij} P_{ii}}{\sum_{j=0}^k P_{ij} + \sum_{j=0}^k P_{ji} - P_{ii}} \quad (5)$$

4.2.2. Computational efficiency metrics

In semantic segmentation, particularly for real-time applications, it is crucial to assess the efficiency of models. Two key metrics commonly used to measure computational complexity are the number of model parameters and Floating Point Operations (FLOPs). These metrics offer insights into the resource demands of a model, with larger parameter counts and higher FLOP values typically indicating greater computational requirements and potentially lower implementation efficiency. Importantly, these metrics are independent of the specific implementation environment, providing a consistent measure of model complexity across different platforms. Additionally, understanding a model's efficiency encompasses both its computational demands and its processing speed. High efficiency is essential for real-time applications, where the balance between accuracy and speed can significantly impact the practical usability of the model.

Frames Per Second (FPS): FPS is a critical metric, particularly in real-time applications like autonomous driving, where processing speed is paramount. FPS quantifies how many frames a model can

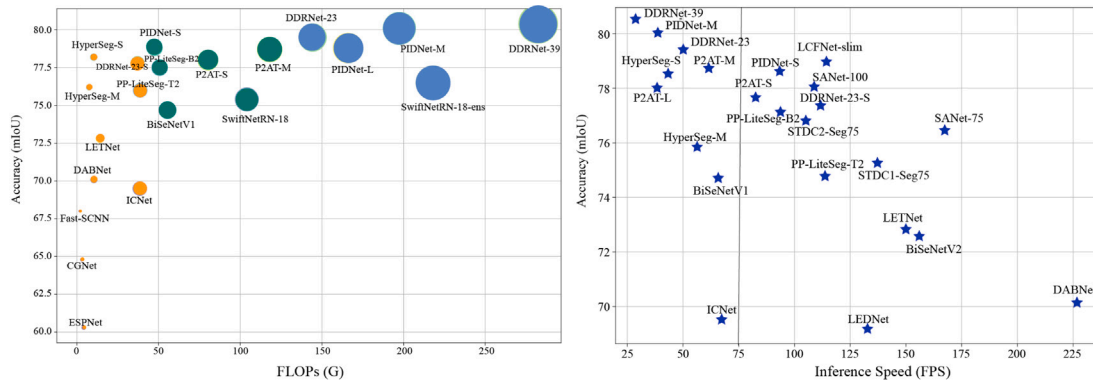


Fig. 11. Comparison of models based on accuracy and speed.

process within a single second, providing a direct measure of its temporal efficiency. High FPS is essential for maintaining responsiveness in dynamic environments, and this metric is often used to benchmark the performance of different segmentation models under real-time constraints.

Model Size (Number of Parameters): The number of trainable parameters serves as a direct indicator of the model's memory requirements and computational footprint. Models with fewer parameters are generally preferred for deployment on embedded systems, where memory is limited.

Floating Point Operations per Second: FLOPs quantify the computational complexity of a model by measuring the total number of floating-point operations required during a single forward pass. This metric is crucial for understanding the trade-off between model accuracy and computational cost, particularly in real-time applications.

Memory Footprint (RAM/VRAM Usage): Efficient memory usage is critical in constrained environments, such as autonomous vehicles, where onboard computational resources are limited. The memory footprint is assessed by the amount of RAM or VRAM utilized during inference, influencing both latency and power consumption.

Latency: Latency refers to the time delay between receiving input data and producing output predictions. Low-latency models are essential for real-time systems, where decisions must be made rapidly to ensure safety and responsiveness.

Energy Consumption: For deployment on mobile platforms and embedded devices, the energy efficiency of a model is of paramount importance. We measure the energy consumed during inference to evaluate the model's suitability for real-world applications in autonomous driving.

Frames Per Second (FPS): Throughput is a critical metric in real-time applications, denoting the number of frames a model can process per second. High FPS values are indicative of models capable of operating under stringent real-time constraints, a necessity for autonomous driving systems where timely scene interpretation is vital.

5. Benchmark analysis

In this section, we evaluate the performance of recent advancements in CNN and transformer-based real time segmentation that been discussed earlier. We have performed analysis on the most commonly used datasets, including Cityscapes Section 5.1, CamVid Section 5.2, ADE20K Section 5.3, and COCO-Stuff Section 5.4, as benchmarks. Additionally, in Sections 5.1 and 5.2, we conduct re-evaluations of several key works, focusing on metrics such as frames per second (FPS), floating-point operations per second (FLOPs), and CPU runtime on an RTX 3090 GPU.

5.1. Benchmark results on cityscapes dataset

The performance analysis of recent state-of-the-art (SOTA) methods on the Cityscapes test dataset provides critical insights into the trade-offs between segmentation accuracy, computational complexity, and real-time processing capabilities key factors for deployment in autonomous vehicles. Table 3 and Fig. 11 show a comparisons and summary of the performance based on different real-time semantic segmentation metrics, such as FPS, FLOPs, and accuracy.

Transformer-based architectures such as SeaFormer (Wan et al., 2023) and LETNet (Xu et al., 2023a) have harnessed the global context effectively to improve performance in the semantic segmentation task. SeaFormer, despite its lightweight design, manages to maintain a competitive mIoU, showcasing the potential of Transformer models in achieving high accuracy with reduced computational overhead. However, the P2AT (Elhassan et al., 2024) variants, particularly P2AT-M and P2AT-L, surpass other models by offering higher mIoU scores, but at the cost of increased computational requirements and inference time. This highlights the inherent trade-offs between model complexity and performance in dense prediction tasks. Very small models such as ENet, ERFNet and LEDNet have proven effective with computational efficiency, while having lower segmentation accuracy. Notably, methods such as SwiftNet (Orsic et al., 2019) and PP-Lite illustrate the feasibility of achieving real-time processing without substantial sacrifices in accuracy, making them suitable for real-world applications where both speed and precision are important. Furthermore, methods like PIDNet (Xu et al., 2023b) and DDRNet (Hong et al., 2021) highlight the shift towards architectures that prioritize real-time performance while maintaining high segmentation accuracy. These models, with their optimized balance of FLOPs and parameters, represent the next frontier in semantic segmentation, where the goal is to achieve state-of-the-art performance in scenarios that demand low latency and efficient computation. Thus, the landscape of semantic segmentation is evolving towards architectures that not only achieve high mIoU scores but also cater to the rigorous requirements of real-time applications in autonomous driving. Ongoing research indicates a promising path towards models that can seamlessly integrate into resource-constrained environments, pushing the limits of both accuracy and efficiency.

Per-class Analysis on Cityscapes Dataset: We provide an analysis of the per-class performance of various real-time semantic segmentation methods evaluated on the Cityscapes test set. The detailed results are presented in Table 4, which highlights the Intersection over Union (IoU) scores across different classes, ranging from road and sidewalk to more challenging categories such as pole, traffic sign, and motorbike.

Best-Performing Classes: Consistent with expectations, most methods demonstrate high performance in classes that typically cover large, homogeneous regions in the scene. For instance, the road, car, and sky classes show some of the highest IoU scores across nearly all methods, with methods such as RELAXNet (Liu et al., 2022a), DMA-Net (Weng

Table 3

Comparison between the other SOTA methods on the Cityscapes test dataset.

Method	work Ref	Year	Backbone	Pretrain	Resolution	GPU	Params (M)	FLOPS (G)	Speed(FPS)	CPU (ms)	Val(mIoU)	Test(mIoU)
SeaFormer-S	Wan et al. (2023)	2023	–	ImageNet	1024 × 1024	GTX 1080Ti	–	8.0	–	–	76.1	75.9
SeaFormer-B	Wan et al. (2023)	2023	–	ImageNet	1024 × 1024	GTX 1080Ti	–	13.7	–	–	77.7	77.5
SegFormer	Xie et al. (2021)	2021	MIT-B0	ImageNet	1024 × 1024	Tesla V100	3.8	125.5	15.2	–	–	76.2
AFFormer-tiny	Dong et al. (2023a)	2023			1024 × 2048		1.6	23.0	–	–	–	76.5
AFFormer-base	Dong et al. (2023a)	2023			1024 × 2048		3.0	34.4	–	–	–	78.7
HSB-Net	Li et al. (2021b)	2021	ResNet-34	ImageNet	512 × 1024	RTX 2080 Ti	12.10	–	124	–	–	73.1
SQNet	Tremi et al. (2016)	2016	SqueezeNet	ImageNet	1024 × 2048	–	–	270.0	17	–	–	59.8
FPANet-C	Wu et al. (2022a)	2022	ResNet18	ImageNet	1024 × 1024	–	14.10	–	–	–	–	75.9
GAS	Lin et al. (2020)	2020	No	No	769 × 1537	Titan Xp	–	–	108.4	–	–	71.8
CFPNet	Lou and Loew (2021)	2021	No	No	1024 × 2048	2080Ti	0.55	–	30	–	–	70.1
LBN-AA	Dong et al. (2020)	2020			360 × 480	Titan X	6.2	49.5	51.0	–	–	73.6
DFANet-A	Li et al. (2019a)	2016	XceptionA	ImageNet	1024 × 1024	Titan X	7.8	3.4	100	–	–	71.3
DFANet-B	Yu and Koltun (2016)	2016	XceptionB	ImageNet	1024 × 1024	Titan X	4.8	2.1	120	–	–	67.1
ANet-16 L	Song et al. (2024)	2024	ANet	ANet	512 × 1024	RTX 3060	13.4	56.4	140.7	–	76.4	76.2
MRDNet	Gu et al. (2024)	2023		ImageNet	1024 × 1024	RTX 2080Ti	8.2	15.	93	–	–	73.1
CSRNet-medium	Xiong et al. (2023)	2023	ResNet-18	ImageNet	768 × 768	GTX 1080Ti	–	–	52.5	–	76.6	75.3
CSRNet-heavy	Xiong et al. (2023)	2023	ResNet-18	ImageNet	768 × 768	GTX 1080Ti	–	–	36.3	–	77.3	76.0
RAFNet	Chen et al. (2023)	2023	–	–	1024 × 2048	RTX A6000	0.96	16.59	107.0	–	75.7	75.3
LCFNet-slim	Yang et al. (2024)	2024		ImageNet	1024 × 2048	RTX 2080Ti	7.6	45.6	114.3	–	78.96	78.95
LCFNet-super	Yang et al. (2024)	2024		ImageNet	1024 × 2048	RTX 2080 Ti	28.27	176.6	47.46	–	80.32	80.3
Light-HRNet-18	Yu et al. (2021b)	2021	Light-HRNet	No	512 × 1024	NVIDIA V100	1.1	1.95	–	–	73.8	72.8
Light-HRNet-30	Yu et al. (2021b)	2021	Light-HRNet	No	512 × 1024	NVIDIA V100	1.8	3.02	–	–	76.0	75.3
DSANet	Elhassan et al. (2021b)	2021	No	No	512 × 1024	RTX3090	3.1	38.0	102.5	193.4	–	71.4
S ² -FPN18	Elhassan et al. (2022b)	2022	ResNet18	ImageNet	512 × 1024	RTX3090	21.6	29.1	179.3	128.9	76.9	76.7
S ² -FPN34	Elhassan et al. (2022b)	2022	ResNet34	ImageNet	512 × 1024	RTX3090	31.7	48.5	135.1	182.0	77.7	77.4
S ² -FPN34M	Elhassan et al. (2022b)	2022	ResNet34	ImageNet	512 × 1024	RTX3090	31.7	190.1	42.6	752.6	78.3	78.2
FSSNet	Zhang et al. (2018a)	2018	No	No	512 × 1024	RTX3090	0.20	4.0	233.2	50.7	–	58.8
ESPNet	Mehta et al. (2018)	2018	ESPNet		512 × 1024	RTX3090	0.36	4.2	330.2	48.9	–	60.3
FPENet	Liu and Yin (2019)	2018	No	No	768 × 1536	RTX3090	0.4	3.5	141.3	147.8	–	70.1
Fast-SCNN	Poudel et al. (2019)	2019			512 × 1024	RTX3090	1.13	1.76	699.3	51.5	–	68.0
SegNet	Badrinarayanan et al. (2017)	2017	VGG-16	ImageNet	360 × 640	RTX3090	29.50	143.1	70	498.7	–	57.0
ERFNet	Romera et al. (2017)	2017	No	No	512 × 1024	RTX3090	2.10	30.1	121.71	1378.	–	68.0
CGNet	Wu et al. (2018)	2020	No	No	360 × 640	RTX3090	0.50	3.1	281.5	30.8	–	64.8
ENet	Paszke et al. (2016)	2015	No	No	512 × 1024	RTX3090	0.36	3.8	135.4	76.4	–	58.3
ICNet	Zhao et al. (2018)	2018	PSPNet50	ImageNet	1024 × 2048	RTX3090	26.5	38.5	67.3	408.8	–	69.5
LEDNet	Wang et al. (2019)	2019		No	512 × 1024	RTX3090	0.76	11.5	132.8	99.9	–	69.2
DABNet	Li et al. (2019b)	2019	DABNet	No	512 × 1024	RTX3090	0.76	10.4	226.6	81.0	–	70.1
BiSeNetV1	Yu et al. (2018b)	2018	ResNet-18	ImageNet	768 × 1536	RTX3090	49.0	55.3	65.5	124.17	74.8	74.7
BiSeNetV2	Yu et al. (2021)	2021	No	No	512 × 1024	RTX3090	–	–	156	166.79	73.4	72.6
SwiftNetRN-18	Orsic et al. (2019)	2019		ImageNet	1024 × 2048	RTX3090	11.8	104.0	39.9	568.08	75.5	75.4
SwiftNetRN-18 ens	Orsic et al. (2019)	2019		ImageNet	1024 × 2048	RTX3090	24.7	218.0	18.4	893.50	–	76.5
SANet-75	Li et al. (2023b)	2023	ResNet-18	ImageNet	768 × 1536	RTX 3090	–	–	167.3	–	77.6	76.6
SANet-100	Li et al. (2023b)	2023	ResNet-18	ImageNet	1024 × 2048	RTX 3090	–	–	109.0	–	79.1	78.1
STDC1-Seg75	Fan et al. (2021)	2021	STDC1	ImageNet	768 × 1536	RTX 3090	8.4	–	130.1	132.30	74.5	75.3
STDC2-Seg75	Fan et al. (2021)	2021	STDC2	ImageNet	768 × 1536	RTX 3090	12.5	–	105.3	184.80	77.0	76.8
PP-LiteSeg-T2	Peng et al. (2022)	2022	STDC1	ImageNet	768 × 1536	RTX 3090	11.9	38.7	113.5	250.08	76.0	74.9
PP-LiteSeg-B2	Peng et al. (2022)	2022	STDC2	ImageNet	768 × 1536	RTX 3090	13.4	50.62	94.2	312.75	78.2	77.5
HyperSeg-M	Nirkin et al. (2021)	2021	EfficientNet-B	ImageNet	512 × 1024	RTX 3090	10.1	7.5	56.4	398.24	76.2	75.8
HyperSeg-S	Nirkin et al. (2021)	2021	EfficientNet-B	ImageNet	768 × 1536	RTX 3090	12	10.27	44.7	635.40	78.2	78.1
PIDNet-S	Xu et al. (2023b)	2023		ImageNet	1024 × 2048	RTX 3090	7.6	47.6	93.2	–	78.8	78.6
PIDNet-M	Xu et al. (2023b)	2023		ImageNet	1024 × 2048	RTX 3090	34.4	197.4	39.8	–	80.1	80.1
DDRNet-23-S	Hong et al. (2021)	2021		ImageNet	1024 × 2048	RTX 3090	5.7	37.1	107.2	174.10	77.8	77.4
DDRNet-23	Hong et al. (2021)	2021		ImageNet	1024 × 2048	RTX 3090	20.1	144.3	50.1	634.52	79.5	79.4
DDRNet-39	Hong et al. (2021)	2021	No	No	1024 × 2048	RTX 3090	32.3	282.4	29.6	1018.59	–	80.4
P2AT-S	Elhassan et al. (2024)	2024	ResNet-18	ImageNet	1024 × 1024	RTX 3090	12.6	80.2	80.1	334.86	78.0	77.8
P2AT-M	Elhassan et al. (2024)	2024	ResNet-34	ImageNet	1024 × 1024	RTX 3090	22.7	118	61.8	476.42	79.8	78.7
P2AT-L	Elhassan et al. (2024)	2024	ResNet-50	ImageNet	1024 × 1024	RTX 3090	37.5	166	40.6	779.76	78.8	78.0
LETNet	Xu et al. (2023a)	2023		No	512 × 1024	RTX 3090	0.95	14.1	71.7	196.3	–	72.8

Table 4

The Per-class, class, and category IoU evaluation on the Cityscapes Test set. List of classes from left to right: Road, Sidewalk, Building, wall, Fence, Pole, Traffic light, Traffic sign, Vegetation, Terrain, Sky, Person, Rider, Car, Truck, Bus, Train, Motor, Bicyclist, mIoU. “cla->mIoU”. “-” indicates the corresponding result is not reported by the methods.

Method	Road	Sidewalk	Building	Wall	Fence	Pole	Traffic light	Traffic sign	Vegetation	Terrain	Sky	Person	Rider	Car	Truck	Bus	Train	Motor	Bicyclist	mIoU
FCN (Long et al., 2015)	97.4	78.4	89.2	34.9	44.2	47.4	60.1	65.0	91.4	69.3	93.9	77.1	51.4	92.6	35.3	48.6	46.5	51.6	66.8	65.3
DMA-Net (Weng et al., 2022)	98.5	85.5	92.2	53.3	55.3	62.5	70.9	74.9	93.0	71.2	95.0	84.0	66.6	95.6	68.2	82.8	76.6	64.5	73.2	77.0
SANet (Li et al., 2023b)	98.4	84.9	92.2	52.9	56.3	63.2	69.5	73.7	92.9	70.5	94.8	83.4	65.9	95.4	71.3	82.6	74.2	61.2	72.0	76.6
SANet (Li et al., 2023b)	98.5	85.5	92.8	54.6	58.0	65.8	72.9	76.4	93.4	71.2	95.3	85.2	68.6	95.7	69.5	82.6	81.1	63.3	74.3	78.1
P2AT-S (Elhassan et al., 2024)	98.4	84.4	92.9	54.4	56.6	67.7	75.2	78.1	93.5	71.9	95.6	85.7	69.2	95.5	62.2	81.1	76.3	64.7	74.9	77.8
P2AT-M (Elhassan et al., 2024)	98.5	85.7	93.0	58.6	58.3	67.3	74.7	77.7	93.5	72.2	95.6	85.3	69.2	95.6	68.1	83.0	78.3	65.9	75.1	78.7
P2AT-L (Elhassan et al., 2024)	98.5	85.3	92.6	53.1	57.4	66.6	74.7	78.1	93.3	69.8	95.1	86.0	70.5	95.8	67.8	83.3	72.2	67.0	74.8	78.0
ESPNet (Mehta et al., 2019)	97.3	78.6	88.8	43.5	42.1	49.3	52.6	60.0	90.5	66.8	93.3	72.9	53.1	91.8	53.0	65.9	53.2	44.2	59.9	60.3
CGNet (Wu et al., 2018)	97.7	81.0	89.8	42.5	48.0	56.2	59.8	65.3	91.4	68.2	94.2	76.8	57.1	92.8	50.8	60.1	51.8	47.3	61.7	64.8
CFPNet (Lou and Loew, 2021)	97.8	81.4	90.5	46.4	50.6	56.4	61.5	67.7	92.1	68.9	94.3	80.4	60.7	93.9	51.4	68.0	50.8	51.2	67.7	70.1
RELAXNet (Liu et al., 2022a)	98.9	84.9	92.2	57.2	54.8	64.3	70.6	74.0	93.0	71.8	94.8	83.7	64.4	95.1	58.6	72.7	58.2	59.9	71.8	74.8
DSANet (Elhassan et al., 2021b)	96.8	78.5	91.2	50.5	50.8	59.4	64.0	71.7	92.6	70.0	93.5	81.8	61.9	92.9	56.1	72.6	50.6	50.9	66.8	71.4
FDWWNet (Liu et al., 2020b)	98.0	82.4	91.1	52.5	51.2	59.9	64.4	68.9	92.5	70.3	94.4	80.8	59.8	94.0	56.5	68.9	48.6	55.7	67.7	71.5
RefineNet (Lin et al., 2017)	97.9	81.3	90.3	48.8	47.4	49.6	57.9	67.3	91.9	69.4	94.2	79.8	59.8	93.7	56.5	67.5	57.5	57.7	68.8	73.6
LEDNet (Wang et al., 2019)	98.1	79.5	91.6	47.7	49.9	62.8	61.3	72.8	92.6	61.2	94.9	76.2	53.7	90.9	64.4	64.0	52.7	44.4	71.6	70.6
AGLNet (Zhou et al., 2020)	97.8	80.1	91.0	51.3	50.6	58.3	63.0	68.5	92.3	71.3	94.2	80.1	59.6	93.8	48.4	68.1	42.1	52.4	67.8	70.1
HAFormer (Xu et al., 2024)	98.4	82.6	91.7	57.2	61.1	63.0	62.2	74.3	91.8	61.7	93.8	79.3	56.4	93.7	66.9	80.1	65.3	56.7	73.5	74.2
LETNet (Xu et al., 2023a)	98.2	83.6	91.6	50.9	53.7	61.0	66.7	70.5	92.5	70.5	94.9	82.3	61.7	94.4	55.0	72.4	57.0	56.1	69.3	72.8

et al., 2022), and P2AT-L (Elhassan et al., 2024) achieving IoUs above 95% for these categories. The superior performance in these classes can be attributed to the relatively straightforward structure and consistent appearance of these elements in urban environments, allowing the models to generalize effectively.

Under-performing Classes: On the other hand, certain classes display notably lower IoU scores, reflecting the inherent difficulty of accurately segmenting these categories. Classes such as pole, traffic light, and motorbike consistently under-perform across various methods. For example, even the more advanced models like SANet (Li et al., 2023b) and HAFormer (Xu et al., 2024) only achieve IoU scores around 65% for the pole class, while performance drops further for the motorbike class. These results suggest that the small size, complex shapes, and variability in appearance of these objects pose significant challenges to the current state-of-the-art segmentation networks. Additionally, the train and rider classes are also among the more challenging categories, often yielding lower IoU scores, which may be due to their less frequent occurrence and varying visual contexts in the dataset.

The analysis underscores a clear dichotomy between the performance on well-represented, homogeneous classes and the challenges faced by models in handling smaller, more complex, and less frequent categories. **This performance gap suggests potential areas for future research**, such as the development of specialized modules or data augmentation techniques to improve segmentation accuracy for underrepresented and difficult classes, thereby enhancing the overall robustness and generalization of real-time semantic segmentation models in diverse real-world scenarios.

5.2. Benchmark results on Camvid dataset

The benchmark results on the Camvid dataset as illustrated in Table 5, highlight a diverse range of state-of-the-art (SOTA) methods, each with varying trade-offs between segmentation accuracy, computational efficiency, and real-time processing capabilities. *The performance comparison reveals distinct categories of models:* lightweight architectures optimized for speed, and more complex networks designed for higher accuracy. Notably, methods like DABNet (Li et al., 2019b) and ICNet (Zhao et al., 2018) emphasize speed with minimal computational overhead, achieving over 200 FPS on an RTX 3090, though with lower mIoU. These models are tailored for scenarios where inference speed of processing is critical, though they sacrifice some accuracy. On the other end, more recent models like HyperSeg-L (Nirkin et al., 2021) and PIDNet-S (Xu et al., 2023b) push the boundaries of accuracy, achieving mIoU scores exceeding 80%, while balancing reasonable computational demands. The P2AT variants, particularly P2AT-L (Elhassan et al., 2024), emerge as top performers in terms of mIoU, with the highest accuracy of 81.1%. However, this comes at the cost of increased parameters and a drop in processing speed, indicating the trade-off between accuracy and efficiency. Models pre-trained on the Cityscapes dataset generally outperform those without pre-training, underscoring the benefits of transfer learning in semantic segmentation tasks. The PIDNet-S-Wider and P2AT-M exemplify this trend, achieving high mIoU scores while managing to retain competitive inference speeds. Overall, the results indicate that the choice of model depends heavily on the specific application requirements, whether the priority is real-time performance or achieving the highest possible accuracy in semantic segmentation tasks. The landscape of real-time semantic segmentation continues to evolve, with newer models consistently performing better in both accuracy and efficiency on small dataset such as Camvid dataset.

5.3. Benchmark results on ADE20K dataset

The ADE20K dataset serves as an important benchmark for evaluating real-time semantic segmentation models, particularly in the context of autonomous vehicle applications where computational efficiency

and segmentation accuracy are crucial. Table 6 provides a detailed comparison of several state-of-the-art architectures, highlighting key metrics such as model parameters (Params), floating-point operations per second (FLOPs), frames per second (FPS), and mean Intersection over Union (mIoU).

This comparison underscores the diverse strategies employed to balance computational demands with segmentation performance. For example, SegFormer (Xie et al., 2021), leveraging the MiT-B0 backbone, achieves a mIoU of 37.4 with only 3.5M parameters and 8.4 GFLOPs, operating at 50.5 FPS on a Tesla V100. This model exemplifies the potential of lightweight transformer architectures in achieving high efficiency without compromising on accuracy. On the other end of the spectrum, the SeaFormer series (Wan et al., 2023) demonstrates remarkable efficiency, particularly on the Qualcomm Snapdragon platform, with its SeaFormer++ variants maintaining an mIoU above 39 while operating with minimal computational overhead. These results highlight the effectiveness of tailored network designs optimized for resource-constrained environments, such as mobile platforms.

RTFormer-B (Wang et al., 2022) prioritizes accuracy, delivering an impressive mIoU of 42.1, but with higher computational costs. This trade-off between accuracy and efficiency is a recurrent theme in real-time semantic segmentation, with different models catering to specific application needs. The TopFormer series (Zhang et al., 2022) further illustrates the potential of efficient architectures, with TopFormer-T achieving a respectable mIoU of 36.1 using only 1.2 GFLOPs. This performance demonstrates that even with limited computational resources, competitive segmentation accuracy is attainable, making these models suitable for deployment on edge devices.

Collectively, these results reflect the ongoing advancements in real-time semantic segmentation. The continuous refinement of these models illustrates the field's commitment to addressing the dual challenges of efficiency and accuracy, which are critical for the practical deployment of autonomous vehicle systems.

5.4. Benchmark results on COCOStuff dataset

Table 7 presents a detailed comparison of state-of-the-art (SOTA) methods on the COCOStuff-10K dataset, emphasizing their performance in real-time semantic segmentation. The methods compared vary in backbone architecture, resolution, GPU utilization, parameter count, FPS, and mean Intersection over Union (mIoU). For instance, ICNet, utilizing the PSP50 backbone, delivers a balanced trade-off between complexity and performance, achieving an mIoU of 29.1 at 35.7 FPS on a Titan X GPU. BiSeNet, on the other hand, demonstrates versatility with different backbones, where the Xception39 variant, though lightweight with only 5.8 million parameters, shows an mIoU of 22.8. The ResNet-18 variant of BiSeNet, while significantly heavier, offers an improved mIoU of 28.1 at a respectable 63.2 FPS. BiSeNetV2-L, with its optimized architecture, sustains a competitive mIoU of 28.7 while maintaining high efficiency on an RTX 2080Ti GPU. In contrast, traditional architectures like Deeplab-v2, using the VGG-16 backbone, while robust, achieve an mIoU of 24.0, reflecting the trade-offs inherent in older models. AFFormer-B and DDRNet-23, with their more modern design, push the boundaries further. AFFormer-B a Transformer based method achieves a notable mIoU of 35.1, though at a lower resolution of 512×512 , whereas DDRNet-23 exhibits remarkable efficiency with an impressive FPS of 129.2, underscoring the benefits of recent advancements in network optimization. Another Transformer based method, RTFormer-B, a leading performer in this comparison, achieves an mIoU of 35.3, the highest among these real-time methods, while maintaining an exceptional FPS of 143.3, demonstrating the potential of Transformer-based approaches in real-time scenarios. Lastly, BFMNet (Liu et al., 2023) and SeaFormer-B offer competitive alternatives, with SeaFormer-B achieving an mIoU of 34.1 on the Snapdragon platform, highlighting the adaptability of these models to diverse hardware configurations. Thus, this comparative analysis highlights the rapid advancements in real-time semantic segmentation, driven by both CNN-based and Transformer-based methodologies, each offering distinct strengths tailored to specific application scenarios.

Table 5

Evaluation of different SOTA methods on Camvid dataset.

Method	Work Ref	backbone	Resolution	GPU	Params	Speed (FPS)	CPU	mIoU%
ENet	Paszke et al. (2016)	No	360 × 480	RTX3090	.36	235.1	39.8	67.1
ICNet	Zhao et al. (2018)	PSPNet-50	360 × 480	RTX3090	26.5	–	19.1	51.3
DABNet	Li et al. (2019b)	DABNet	360 × 480	RTX3090	0.76	457.0	30.5	66.4
BiSeNet1	Yu et al. (2018b)	ImageNet	720 × 960	RTX3090	5.8	175		65.7
BiSeNet2	Yu et al. (2018b)	ImageNet	720 × 960	RTX3090	49.0	116.3	163.90	68.7
DFANet-A	Li et al. (2019a)	ImageNet	720 × 960	RTX3090	7.8	120	352.36	64.7
DFANet-B	Li et al. (2019a)	ImageNet	720 × 960	RTX3090	4.8	160	300.56	59.3
BiSeNetV2	Yu et al. (2021)	no	720 × 960	RTX3090	–	124.5	242.63	76.7
STDC1-Seg	Fan et al. (2021)	ImageNet	720 × 960	RTX3090	8.4	197.6	144.46	73.0
STDC2-Seg	Fan et al. (2021)	ImageNet	720 × 960	RTX3090	12.5	152.2	185.10	73.9
DDRNet-23-slim	Hong et al. (2021)	No	720 × 960	RTX3090	5.7	–	50.04	74.4
DDRNet-23	Hong et al. (2021)	No	720 × 960	RTX3090	20.0	–	158.73	76.3
HyperSeg-S	Nirkin et al. (2021)	ImageNet	720 × 960	RTX3090	9.9	38.0	210.34	78.4
HyperSeg-L	Nirkin et al. (2021)	ImageNet	720 × 960	RTX3090	10.2	16.6	305.49	79.1
PP-LiteSeg-T2	Peng et al. (2022)	ImageNet	720 × 960	RTX3090	13.4	222.3	124.3	73.3
PP-LiteSeg-B2	Peng et al. (2022)	ImageNet	720 × 960	RTX3090	11.9	154.8	101.0	75.0
CIDNet2-50*	Dong et al. (2023b)	No	720 × 960	RTX3090	7.1	109.6	–	77.8
SANet	Li et al. (2023b)	ImageNet	720 × 960	RTX3090	–	250		77.2
ANet-16 L*	Song et al. (2024)	ImageNet	720 × 960	RTX3090	13.4	118.9	–	79.5
w/ Cityscapes pre-training								
BiSeNetV2-L ^a	Yu et al. (2021)	Cityscapes	720 × 960	RTX3090	32.7	–	–	78.5
DDRNet-23-S ^a	Hong et al. (2021)	Cityscapes	720 × 960	RTX3090	5.7	182.4	–	78.0
DDRNet-23 ^a	Hong et al. (2021)	Cityscapes	720 × 960	RTX3090	20.0	116.8	158.73	79.9
PIDNet-S ^a	Xu et al. (2023b)	Cityscapes	720 × 960	RTX3090	7.6	153.7	–	80.1
PIDNet-S-Wider ^a	Xu et al. (2023b)	Cityscapes	720 × 960	RTX3090	–	85.6	–	82.0
P2AT-S*	Elhassan et al. (2024)	Cityscapes	720 × 960	RTX3090	12.6	113.6	217.61	80.5
P2AT-M*	Elhassan et al. (2024)	Cityscapes	720 × 960	RTX3090	22.7	86.6	323.60	81.0
P2AT-L*	Elhassan et al. (2024)	Cityscapes	720 × 960	RTX3090	37.5	56.1	449.81	81.1

^a Indicates the model pre-trained on Cityscapes.**Table 6**

Popular common real-time semantic segmentation architectures on ADE20K.

Method category	refs	Backbone	GPU	Params	FLOPs	FPS	mIoU
PSPNet	Zhao et al. (2017)	MobileNetV2	Tesla V100	13.7	52.9	57.7	29.6
SegFormer	Xie et al. (2021)	MiT-B0	Tesla V100	3.5	8.4	50.5	37.4
FCN	Long et al. (2015)	MobileNetV2	Tesla V100	9.8	39.6	64.4	19.7
DeepLabV3+	Chen et al. (2018)	MobileNetV2	Tesla V100	15.4	69.4	43.1	34.0
RTFormer-B	Wang et al. (2022)	–	RTX2080Ti	16.8	–	71.4	42.1
SeaFormer-S	Wan et al. (2023)	SeaFormer-S	Qualcomm Snapdrag	4.0	1.1	–	38.1
SeaFormer-S++	Wan et al. (2024)	SeaFormer-S++	Qualcomm Snapdrag	4.1	1.1	–	39.7
SeaFormer-B	Wan et al. (2023)	SeaFormer-B	Qualcomm Snapdrag	8.6	1.8	–	40.2
SeaFormer-B++	Wan et al. (2024)	SeaFormer-B++	Qualcomm Snapdrag	8.7	1.8	–	41.0
TopFormer-T	Zhang et al. (2022)	TopFormer-T	–	3.1	1.2	–	36.1
TopFormer-B	Zhang et al. (2022)	TopFormer-B	–	5.1	1.8	–	37.8

Table 7

Comparison of SOTA methods on COCOStuff-10K dataset.

Method category	refs	Backbone	Resolution	GPU	Paras (M)	FPS	mIoU
ICNet	Zhao et al. (2018)	PSP50	640 × 640	Titan X	26.5	35.7	29.1
BiSeNet	Yu et al. (2018b)	Xception39	640 × 640	Titan X	5.8	–	22.8
BiSeNet	Yu et al. (2018b)	ResNet-18	640 × 640	Titan X	49.0	63.2	28.1
BiSeNetV2-L	Yu et al. (2021)	–	640 × 640	RTX 2080Ti		42.5	28.7
Deeplab-v2	Chen et al. (2017)	VGG-16	640 × 640	Titan X	–	58.2	24.0
AFFormer-B	Dong et al. (2023a)	AFFormer	512 × 512	V100	3.0	–	35.1
DDRNet-23	Hong et al. (2021)	–	640 × 640	RTX 2080Ti	20.1M	129.2	32.1
RTFormer-B	Wang et al. (2022)	–	640 × 640	RTX2080Ti	16.8	143.3	35.3
BFMNet	Liu et al. (2023)	ResNet18	640 × 640	RTX 2080Ti	22.3	101	31.2
SeaFormer-B	Wan et al. (2023)	SeaFormer-B	640 × 640	Qualcomm Snapdrag	8.6	–	34.1

Table 8

Acronyms and descriptions for real-time semantic segmentation in autonomous driving.

Acronym	Description
CNNs	Convolutional Neural Networks
AVs	Autonomous Vehicles
FPS	Frames Per Second
CPU	Central processing unit
GPU	Graphics Processing unit
ML	Machine learning
DL	Deep Learning
ADAS	Advanced Driver Assistance Systems
ACC	Adaptive Cruise Control
LCA	Lateral Guidance Assistance
DCNNs	Deep convolutional neural networks
FCN	Fully Convolutional Network
ViT	Vision Transformer
FLOPS	Floating-point operations per second
MHSA	Multi-head Self-attention
NAS	Neural Architecture Search
PA	Pixel Accuracy
MPA	Mean Pixel Accuracy
IoU	Intersection over Union
mIoU	Mean Intersection over Union
FWIoU	Frequency-Weighted Intersection over Union
RAM	Random Access Memory
VRAM	Video Random Access Memory
SAM	Segment Anything Model
RADAR	Radio Detection and Ranging
LiDAR	Light Detection and Ranging
NLP	Natural Language Processing
COCO	Common Objects in Context-stuff
CamVid	Cambridge-driving Labeled Video Database
FP	False Positive
FN	False Negative
TP	True Positive
MPA	Mean Pixel Accuracy
PA	Pixel Accuracy
FWIoU	Frequency-Weighted Intersection over Union
mIoU	Mean Intersection over Union
IoU	Intersection over Union

6. Challenges and research directions

6.1. Key challenges in semantic segmentation

6.1.1. The trade-off between accuracy and efficiency

Achieving an optimal balance between accuracy and efficiency remains a significant challenge in semantic segmentation. High-accuracy models, such as PSPNet (Zhao et al., 2017) and HRNet (Wang et al., 2020b), typically come at the cost of reduced computational efficiency. In contrast, models that excel in efficiency, like LETNet (Xu et al., 2023a) and ENet (Paszke et al., 2016), often sacrifice segmentation accuracy. Among the current methods, DFANet (Li et al., 2019a) has emerged as a leading approach to balance between these two crucial factors. However, it still falls short in accuracy when compared to top-performing models, indicating a persistent trade-off that needs to be addressed.

6.1.2. The challenge of high-quality training data

For accurate semantic segmentation, High-quality training data is essential for accurate semantic segmentation, yet acquiring such data is labor-intensive and time-consuming. The need for extensive labeled datasets with pixel-level annotations presents a major challenge. To

mitigate this dependency, researchers have explored weakly- and semi-supervised methods that operate under the assumption of limited high-quality data. While these approaches offer some relief, their performance still lags behind fully supervised methods, highlighting an area that requires further investigation.

6.1.3. Addressing domain gaps across datasets

The domain gap is a universal issue in semantic segmentation and other computer vision tasks. For instance, while PSPNet achieves impressive results on the PASCAL VOC 2012 dataset (Everingham et al., 2010) with an 85.4% mIoU, its performance drops significantly to 44.94% MIOU on the more challenging ADE20K dataset (Zhou et al., 2019), which features a greater number of semantic classes and complex scenes. This discrepancy arises because different datasets vary in terms of categories, scene complexity, dataset size, and object scale. These variations contribute to the domain gap, underscoring the importance of addressing this issue when developing and deploying semantic segmentation models across diverse applications.

6.2. Future directions for semantic segmentation

6.2.1. Harnessing transformer for real-time segmentation

As highlighted in Table 3, few real-time semantic segmentation algorithms have adopted Transformer architectures or hybrid Transformer-CNN frameworks as their primary model. Most existing models still rely heavily on CNNs, likely due to CNNs' superior optimization and acceleration when deployed, and their ease of training on smaller datasets. However, Transformers offer powerful context extraction capabilities (Liu et al., 2021a) and flexible mask generation methods (Jain et al., 2023), which have significantly advanced semantic segmentation. The challenge lies in effectively applying the innovations in lightweight Transformer design to enhance real-time semantic segmentation performance.

6.2.2. Expanding beyond traditional datasets

Current real-time semantic segmentation algorithms predominantly focus on datasets related to autonomous driving, such as Cityscapes and Camvid. Expanding research to include more varied and challenging datasets, including general scene segmentation or other specialized segmentation tasks, is essential to further evaluate and improve the generalization and adaptability of these algorithms.

6.2.3. Optimizing real-time segmentation for edge and mobile devices

Most real-time semantic segmentation algorithms are currently optimized for desktop GPUs. However, practical applications often require deployment on mobile and edge devices, which are constrained by limited memory and lower computational power. Achieving a balance between accuracy and speed on these devices remains a significant challenge, as the performance of real-time segmentation networks on mobile and edge devices lags behind that of desktop counterparts. Bridging this gap and ensuring these networks are effectively integrated into specific application domains is a crucial future direction.

6.2.4. Knowledge distillation from large models to lightweight networks

Recent advancements in large-scale visual models such as DINO (Oquab et al., 2023; Caron et al., 2021), SEEM (Zou et al., 2024), and Segment Anything Model (SAM) (Kirillov et al., 2023) have revolutionized various computer vision tasks, including semantic segmentation. SAM's strong zero-shot transfer capabilities have sparked interest in its potential application to autonomous driving, especially in challenging scenarios such as adverse weather conditions. However, integrating SAM's benefits into real-time segmentation tasks, which demand lightweight networks, remains challenging.

One promising solution is knowledge distillation, where the strengths of large models like SAM are transferred to smaller, more efficient networks. For instance, Mobile-SAM (Zhang et al., 2023b,a)

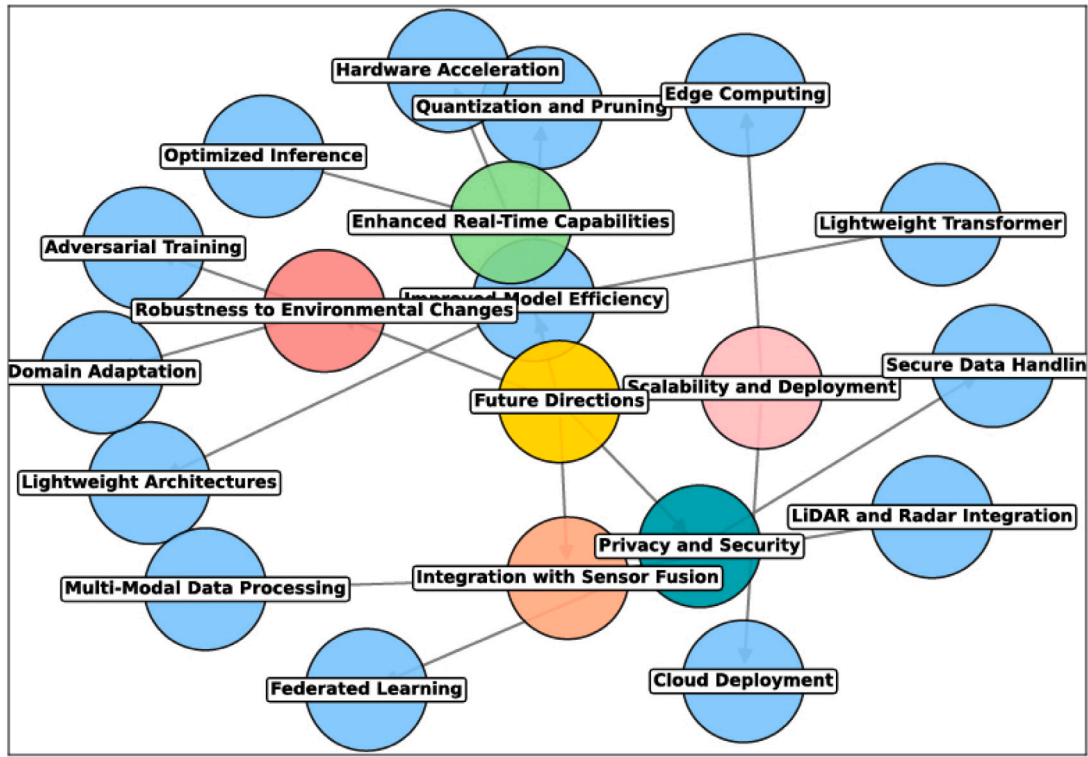


Fig. 12. Future Directions in Real-Time Semantic Segmentation for Autonomous Vehicles.

and Fast-SAM (Zhao et al., 2023) have demonstrated the feasibility of this approach, creating lightweight models that retain much of SAM's capability while being suitable for deployment on mobile and edge devices. This future direction emphasizes the need to further explore and refine such distillation techniques, particularly focusing on SAM's robustness in adverse conditions, to enhance its applicability in real-time autonomous driving systems

6.2.5. Scalability of real-time segmentation models

As real-time segmentation models are deployed in increasingly diverse environments, ensuring scalability across different hardware platforms, from high-end GPUs to low-power edge devices, becomes critical. Future research should focus on creating models that can dynamically scale their complexity based on available computational resources while maintaining performance.

6.2.6. Robustness to adverse conditions

Real-time semantic segmentation models need to perform reliably under various adverse conditions, such as poor lighting, weather changes, occlusions, and motion blur. Developing methods that enhance the robustness of these models in such scenarios, possibly through data augmentation techniques or robustness-focused architectures, is an essential direction for future work.

6.2.7. Energy-efficient inference

As edge computing and mobile applications grow, the energy consumption of real-time segmentation models becomes a critical concern. Future research could explore energy-efficient model designs, including hardware-aware neural architecture search (NAS) (Ren et al., 2021; Liu et al., 2021b, 2018; Pham et al., 2018) and low-power hardware optimizations, to reduce the energy footprint of these models without sacrificing performance (Daghero et al., 2021).

6.2.8. Multi-modal segmentation

Incorporating additional sensor data, such as LiDAR (Liu et al., 2024), RADAR (Prophet et al., 2019; Yao et al., 2023a), or thermal imaging (Sun et al., 2020; Kütük and Algan, 2022), could enhance the accuracy and reliability of real-time semantic segmentation, especially in challenging environments. Developing multi-modal segmentation approaches that can effectively fuse information from different sensors in real-time is a promising area for future exploration.

6.2.9. Ethical and bias considerations

As real-time segmentation models are increasingly used in critical applications such as autonomous driving, ensuring that these models are free from bias and operate ethically is paramount (Islam, 2024; Kriebitz et al., 2022; Cunneen et al., 2019, 2020). Future research should focus on developing methods for detecting and mitigating biases in training data and model predictions, ensuring fair and equitable performance across different demographic groups and environmental conditions.

7. Conclusion

Real-time semantic segmentation is a cornerstone in the realm of autonomous driving, playing a critical role in the vehicle's perception and decision-making processes. The demand for efficient, and high inference speed algorithms is ever-growing, driven by the need for instantaneous response times within constrained computational environments. This survey has provided a comprehensive analysis of the advancements in real-time semantic segmentation, highlighting the evolution from traditional CNN-based approaches to the more recent Transformer-based models. We have systematically examined the methodologies that have been developed to balance accuracy with computational efficiency, particularly within the context of autonomous vehicles. Our discussion has covered key areas such as architectural innovations, and the integration of lightweight modules that enhance both speed and performance. By dissecting these approaches from the perspective of network architecture, module design, and practical

application scenarios, we have offered a clear view of the current landscape and the potential paths forward. Despite the remarkable progress, challenges remain, particularly in achieving higher levels of precision without compromising real-time capabilities. Future research must address these challenges, exploring novel strategies that further optimize the speed-accuracy trade-off while adapting to the evolving demands of autonomous driving. We have also provided a detailed overview of the datasets and evaluation metrics commonly used in this domain, offering a valuable resource for researchers aiming to benchmark their models effectively. Additionally, our quantitative assessments of existing algorithms serve as a crucial guide for those seeking to understand the practical implications of deploying these technologies in real-world scenarios. Furthermore, we have evaluated several key algorithms using unified device, RTX 3090 to benchmark the inference speeds, Flops, and CPU run time. Thus, this survey not only maps the current state of real-time semantic segmentation but also overlays the way for future innovations that will drive the next generation of autonomous vehicles. The insights and directions provided herein are intended to inspire continued advancements in this critical area of research, ultimately contributing to safer, more efficient autonomous driving systems.

The algorithms, datasets, and evaluation metrics referenced in this article are comprehensively summarized in a dedicated GitHub repository. Readers can access these resources at.⁴

CRedit authorship contribution statement

Mohammed A.M. Elhassan: Writing – original draft, Visualization, Methodology, Conceptualization. **Changjun Zhou:** Supervision, Resources, Project administration, Funding acquisition. **Ali Khan:** Writing – review & editing, Validation. **Amina Benabid:** Visualization, Formal analysis. **Abuzar B.M. Adam:** Writing – review & editing, Validation. **Atif Mehmood:** Writing – review & editing. **Naftaly Wambugu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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⁴ <https://github.com/mohamedac29/Real-time-Semantic-Segmentation-Survey>

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